NCERT Chemistry
Complete Exercise Book

Class XI and Class XII

566 exercises across 19 chapters

Class XI: 323 | Class XII: 243

Class XI Chemistry

Unit 1: Some Basic Concepts of Chemistry

34 exercises

Q 1.1
Calculate the molar mass of the following:
(i) H2O (ii) CO2 (iii) CH4

Q 1.2
Calculate the mass per cent of different elements present in sodium sulphate (Na2SO4).

Q 1.3
Determine the empirical formula of an oxide of iron, which has 69.9% iron and 30.1% dioxygen by mass.

Q 1.4
Calculate the amount of carbon dioxide that could be produced when

Q 1.5
(i) 1 mole of carbon is burnt in air. (ii) 1 mole of carbon is burnt in 16 g of dioxygen. (iii) 2 moles of carbon are burnt in 16 g of dioxygen. Calculate the mass of sodium acetate (CH3COONa) required to make 500 mL of 0.375 molar aqueous solution. Molar mass of sodium acetate is 82.0245 g mol-1. 26

Q 1.6
Calculate the concentration of nitric acid in moles per litre in a sample which has a density, 1.41 g mL-1 and the mass per cent of nitric acid in it being 69%.

Q 1.7
How much copper can be obtained from 100 g of copper sulphate (CuSO4)?

Q 1.8
Determine the molecular formula of an oxide of iron, in which the mass per cent of iron and oxygen are 69.9 and 30.1, respectively.

Q 1.9
Calculate the atomic mass (average) of chlorine using the following data:

Q 1.10
% Natural Abundance Molar Mass 35 Cl 75.77 34.9689 37 Cl 24.23 36.9659 In three moles of ethane (C2H6), calculate the following: (i) Number of moles of carbon atoms. (ii) Number of moles of hydrogen atoms. (iii) Number of molecules of ethane.

Q 1.11
What is the concentration of sugar (C12H22O11) in mol L-1 if its 20 g are dissolved in enough water to make a final volume up to 2L?

Q 1.12
If the density of methanol is 0.793 kg L-1, what is its volume needed for making 2.5 L of its 0.25 M solution?

Q 1.13
Pressure is determined as force per unit area of the surface. The SI unit of pressure, pascal is as shown below: 1Pa = 1N m-2 If mass of air at sea level is 1034 g cm-2, calculate the pressure in pascal.

Q 1.14
What is the SI unit of mass? How is it defined?

Q 1.15
Match the following prefixes with their multiples: Prefixes Multiples (i) micro (ii) deca (iii) mega 10-6 (iv) giga 10-15 (v) femto 10

Q 1.16
What do you mean by significant figures?

Q 1.17
A sample of drinking water was found to be severely contaminated with chloroform, CHCl3, supposed to be carcinogenic in nature. The level of contamination was 15 ppm (by mass).

Q 1.18
1.19 (i) Express this in per cent by mass. (ii) Determine the molality of chloroform in the water sample. Express the following in the scientific notation: (i) 0.0048 (ii) 234,000 (iii) 8008 (iv) 500.0 (v) 6.0012 How many significant figures are present in the following? (i) 0.0025 (ii) (iii) 5005 Some Basic Concepts of Chemistry

Q 1.20

1.21 (iv) 126,000 (v) 500.0 (vi) 2.0034 27 Round up the following upto three significant figures: (i) 34.216 (ii) 10.4107 (iii) 0.04597 (iv) 2808 The following data are obtained when dinitrogen and dioxygen react together to form different compounds: Mass of dinitrogen Mass of dioxygen (i) 14 g 16 g (ii) 14 g 32 g
(iii) 28 g 32 g
(iv) 28 g 80 g (a) Which law of chemical combination is obeyed by the above experimental data? Give its statement.
(b) Fill in the blanks in the following conversions: (i) 1 km = ... mm = ... pm (ii) 1 mg = ... kg = ... ng (iii) 1 mL = ... L = ... dm3

Q 1.22

If the speed of light is 3.0 x 108 m s-1, calculate the distance covered by light in 2.00 ns.

Q 1.23

In a reaction A + B2 d AB2 Identify the limiting reagent, if any, in the following reaction mixtures.

Q 1.24

(i) 300 atoms of A + 200 molecules of B (ii) 2 mol A + 3 mol B (iii) 100 atoms of A + 100 molecules of B (iv) 5 mol A + 2.5 mol B (v) 2.5 mol A + 5 mol B Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation: N2 (g) + H2 (g) d 2NH3 (g) (i) Calculate the mass of ammonia produced if 2.00 x 103 g dinitrogen reacts with 1.00 x 103 g of dihydrogen. (ii) Will any of the two reactants remain unreacted? (iii) If yes, which one and what would be its mass?

Q 1.25

How are 0.50 mol Na2CO3 and 0.50 M Na2CO3 different?

Q 1.26

If 10 volumes of dihydrogen gas reacts with five volumes of dioxygen gas, how many volumes of water vapour would be produced?

Q 1.27

Convert the following into basic units: (i) 28.7 pm (ii) 15.15 pm (iii) 25365 mg 28

Q 1.28

Which one of the following will have the largest number of atoms? (i) 1 g Au (s) (ii) 1 g Na (s) (iii) 1 g Li (s) (iv) 1 g of Cl2(g)

Q 1.29

Calculate the molarity of a solution of ethanol in water, in which the mole fraction of ethanol is 0.040 (assume the density of water to be one).

Q 1.30

What will be the mass of one 12C atom in g?

Q 1.31

How many significant figures should be present in the answer of the following calculations? 0.02856 x 298.15 x 0.112 0.5785 (iii) 0.0125 + 0.7864 + 0.0215 (ii) (i)

Q 1.32

5 x 5.364 Use the data given in the following table to calculate the molar mass of naturally occuring argon isotopes: Isotope Isotopic molar mass Abundance 36 Ar 35.96755 g mol-1 0.337% 38 Ar -1 37.96272 g mol 0.063% 40 Ar -1 39.9624 g mol 99.600%

Q 1.33

Calculate the number of atoms in each of the following (i) 52 moles of Ar (ii) 52 u of He
(iii) 52 g of He.

Q 1.34

A welding fuel gas contains carbon and hydrogen only. Burning a small sample of it in oxygen gives 3.38 g carbon dioxide, 0.690 g of water and no other products. A volume of 10.0 L (measured at STP) of this welding gas is found to weigh 11.6 g. Calculate
(i) empirical formula, (ii) molar mass of the gas, and (iii) molecular formula.

Q 1.35

Calcium carbonate reacts with aqueous HCl to give CaCl2 and CO2 according to the reaction, CaCO3 (s) + 2 HCl (aq) -> CaCl2 (aq) + CO2(g) + H2O(l) What mass of CaCO3 is required to react completely with 25 mL of 0.75 M HCl?

Q 1.36
Chlorine is prepared in the laboratory by treating manganese dioxide (MnO2) with aqueous hydrochloric acid according to the reaction 4 HCl (aq) + MnO2(s) -> 2H2O (l) + MnCl2(aq) + Cl2 (g) How many grams of HCl react with 5.0 g of manganese dioxide?

Unit 2: Structure of Atom

68 exercises

Q 2.1
(i) Calculate the number of electrons which will together weigh one gram.
(ii) Calculate the mass and charge of one mole of electrons.

Q 2.2
(i) Calculate the total number of electrons present in one mole of methane.
(ii) Find (a) the total number and (b) the total mass of neutrons in 7 mg of 14C. (Assume that mass of a neutron = 1.675 x 10-27 kg).
(iii) Find (a) the total number and (b) the total mass of protons in 34 mg of NH3 at STP.

Q 2.3
How many neutrons and protons are there in the following nuclei ? 13 6

Q 2.4
Will the answer change if the temperature and pressure are changed ? 56 88 C, 168 O, 24 12 Mg, 26 Fe, 38 Sr Write the complete symbol for the atom with the given atomic number (Z) and atomic mass (A) (i) Z = 17, A = 35.
(ii) Z = 92, A = 233.
(iii) Z = 4, A = 9.

Q 2.5
Yellow light emitted from a sodium lamp has a wavelength (lambda) of 580 nm. Calculate the frequency (nu) and wavenumber ( ) of the yellow light.

Q 2.6
Find energy of each of the photons which (i) correspond to light of frequency 3x1015 Hz.
(ii) have wavelength of 0.50 A.

Q 2.7
Calculate the wavelength, frequency and wavenumber of a light wave whose period is 2.0 x 10-10 s.

Q 2.8
What is the number of photons of light with a wavelength of 4000 pm that provide 1J of energy?

Q 2.9
A photon of wavelength 4 x 10-7 m strikes on metal surface, the work function of the metal being 2.13 eV. Calculate (i) the energy of the photon (eV),
(ii) the kinetic energy of the emission, and (iii) the velocity of the photoelectron (1 eV= 1.6020 x 10-19 J).

Q 2.10
Electromagnetic radiation of wavelength 242 nm is just sufficient to ionise the sodium atom. Calculate the ionisation energy of sodium in kJ mol-1.

Q 2.11
A 25 watt bulb emits monochromatic yellow light of wavelength of 0.57um. Calculate the rate of emission of quanta per second.

Q 2.12
Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6800 A. Calculate threshold frequency (nu0) and work function (W0) of the metal.

Q 2.13
What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from an energy level with n = 4 to an energy level with n = 2? 70

Q 2.14
How much energy is required to ionise a H atom if the electron occupies n = 5 orbit? Compare your answer with the ionization enthalpy of H atom (energy required to remove the electron from n =1 orbit).

Q 2.15
What is the maximum number of emission lines when the excited electron of a H atom in n = 6 drops to the ground state?

Q 2.16

(i) The energy associated with the first orbit in the hydrogen atom is -2.18 x 10-18 J atom-1. What is the energy associated with the fifth orbit?

(ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

Q 2.17

Calculate the wavenumber for the longest wavelength transition in the Balmer series of atomic hydrogen.

Q 2.18

What is the energy in joules, required to shift the electron of the hydrogen atom from the first Bohr orbit to the fifth Bohr orbit and what is the wavelength of the light emitted when the electron returns to the ground state? The ground state electron energy is -2.18 x 10-11 ergs.

Q 2.19

The electron energy in hydrogen atom is given by En = (-2.18 x 10-18)/n2 J. Calculate the energy required to remove an electron completely from the n = 2 orbit. What is the longest wavelength of light in cm that can be used to cause this transition?

Q 2.20

Calculate the wavelength of an electron moving with a velocity of 2.05 x 107 m s-1.

Q 2.21

The mass of an electron is 9.1 x 10-31 kg. If its K.E. is 3.0 x 10-25 J, calculate its wavelength.

Q 2.22

Which of the following are isoelectronic species i.e., those having the same number of electrons? Na+, K+, Mg2+, Ca2+, S2-, Ar.

Q 2.23

(i) Write the electronic configurations of the following ions: (a) H- (b) Na+ (c) O2-

(d) F-

(ii) What are the atomic numbers of elements whose outermost electrons are represented by (a) 3s1 (b) 2p3 and (c) 3p5 ?

(iii) Which atoms are indicated by the following configurations ?

(a) [He] 2s1 (b) [Ne] 3s2 3p3 (c) [Ar] 4s2 3d1.

Q 2.24

What is the lowest value of n that allows g orbitals to exist?

Q 2.25

An electron is in one of the 3d orbitals. Give the possible values of n, l and ml for this electron.

Q 2.26

An atom of an element contains 29 electrons and 35 neutrons. Deduce (i) the number of protons and (ii) the electronic configuration of the element.

Q 2.27

Give the number of electrons in the species

Q 2.28

(i) An atomic orbital has n = 3. What are the possible values of l and ml ?

(ii) List the quantum numbers (ml and l) of electrons for 3d orbital.

(iii) Which of the following orbitals are possible? 1p, 2s, 2p and 3f

Q 2.29

Using s, p, d notations, describe the orbital with the following quantum numbers.

(a) n=1, l=0; (b) n = 3; l=1 (c) n = 4; l =2; (d) n=4; l=3.

Q 2.30

Explain, giving reasons, which of the following sets of quantum numbers are not possible. (a) n = 0, l = 0, ml = 0, ms = + 1/2 (b) n = 1, l = 0, ml = 0, ms = - 1/2 (c) n = 1, l = 1, n = 2, l = 1, ml = 0, ms = + 1/2 (d) ml = 0, ms = - 1/2 71 structure of atom (e) n = 3, l = 3, ml = -3, ms = + 1/2 (f) n = 3, l = 1, ml = 0, ms = + 1/2

Q 2.31

How many electrons in an atom may have the following quantum numbers?

(a) n = 4, ms = - 1/2

(b) n = 3, l = 0

Q 2.32
Show that the circumference of the Bohr orbit for the hydrogen atom is an integral multiple of the de Broglie wavelength associated with the electron revolving around the orbit.

Q 2.33
What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 of He+ spectrum ?

Q 2.34
Calculate the energy required for the process He+ (g) g He2+ (g) + e- The ionization energy for the H atom in the ground state is 2.18 x 10-18 J atom-1

Q 2.35
If the diameter of a carbon atom is 0.15 nm, calculate the number of carbon atoms which can be placed side by side in a straight line across length of scale of length 20 cm long.

Q 2.36
2 x108 atoms of carbon are arranged side by side. Calculate the radius of carbon atom if the length of this arrangement is 2.4 cm.

Q 2.37
The diameter of zinc atom is 2.6 A. Calculate (a) radius of zinc atom in pm and
(b) number of atoms present in a length of 1.6 cm if the zinc atoms are arranged side by side lengthwise.

Q 2.38
A certain particle carries 2.5 x 10-16C of static electric charge. Calculate the number of electrons present in it.

Q 2.39
In Milikan's experiment, static electric charge on the oil drops has been obtained by shining X-rays. If the static electric charge on the oil drop is -1.282 x 10-18C, calculate the number of electrons present on it.

Q 2.40
In Rutherford's experiment, generally the thin foil of heavy atoms, like gold, platinum etc. have been used to be bombarded by the alpha-particles. If the thin foil of light atoms like aluminium etc. is used, what difference would be observed from the above results ?

Q 2.41
35 and 79Br can be written, whereas symbols 79 Symbols 79 Br and 35Br are not 35 Br acceptable. Answer briefly.

Q 2.42
An element with mass number 81 contains 31.7% more neutrons as compared to protons. Assign the atomic symbol.

Q 2.43
An ion with mass number 37 possesses one unit of negative charge. If the ion conatins 11.1% more neutrons than the electrons, find the symbol of the ion.

Q 2.44
An ion with mass number 56 contains 3 units of positive charge and 30.4% more neutrons than electrons. Assign the symbol to this ion.

Q 2.45
Arrange the following type of radiations in increasing order of frequency: (a) radiation from microwave oven (b) amber light from traffic signal (c) radiation from FM radio
(d) cosmic rays from outer space and (e) X-rays.

Q 2.46
Nitrogen laser produces a radiation at a wavelength of 337.1 nm. If the number of photons emitted is 5.6 x 1024, calculate the power of this laser.

Q 2.47
Neon gas is generally used in the sign boards. If it emits strongly at 616 nm, calculate (a) the frequency of emission, (b) distance traveled by this radiation in 30 s (c) energy of quantum and (d) number of quanta present if it produces 2 J of energy. 72

Q 2.48
In astronomical observations, signals observed from the distant stars are generally weak. If the photon detector receives a total of 3.15 x 10-18 J from the radiations of 600 nm, calculate the number of photons received by the detector.

Q 2.49
Lifetimes of the molecules in the excited states are often measured by using pulsed radiation source of duration nearly in the nano second range. If the radiation source has the duration of 2 ns and the number of photons emitted during the pulse source is 2.5 x 1015, calculate the energy of the source.

Q 2.50
The longest wavelength doublet absorption transition is observed at 589 and 589.6 nm. Calcualte the frequency of each transition and energy difference between two excited states.

Q 2.51
The work function for caesium atom is 1.9 eV. Calculate (a) the threshold wavelength and (b) the threshold frequency of the radiation. If the caesium element is irradiated with a wavelength 500 nm, calculate the kinetic energy and the velocity of the ejected photoelectron.

Q 2.52
Following results are observed when sodium metal is irradiated with different wavelengths. Calculate (a) threshold wavelength and, (b) Planck's constant. lambda (nm) v x 10-5 (cm s-1)

Q 2.55
4.35 5.35

Q 2.53
The ejection of the photoelectron from the silver metal in the photoelectric effect experiment can be stopped by applying the voltage of 0.35 V when the radiation 256.7 nm is used. Calculate the work function for silver metal.

Q 2.54
If the photon of the wavelength 150 pm strikes an atom and one of tis inner bound electrons is ejected out with a velocity of 1.5 x 107 m s-1, calculate the energy with which it is bound to the nucleus.

Q 2.55
Emission transitions in the Paschen series end at orbit n = 3 and start from orbit n and can be represeted as v = 3.29 x 1015 (Hz) [1/32 - 1/n2] Calculate the value of n if the transition is observed at 1285 nm. Find the region of the spectrum.

Q 2.56
Calculate the wavelength for the emission transition if it starts from the orbit having radius 1.3225 nm and ends at 211.6 pm. Name the series to which this transition belongs and the region of the spectrum.

Q 2.57
Dual behaviour of matter proposed by de Broglie led to the discovery of electron microscope often used for the highly magnified images of biological molecules and other type of material. If the velocity of the electron in this microscope is 1.6 x 106 ms-1, calculate de Broglie wavelength associated with this electron.

Q 2.58
Similar to electron diffraction, neutron diffraction microscope is also used for the determination of the structure of molecules. If the wavelength used here is 800 pm, calculate the characteristic velocity associated with the neutron.

Q 2.59
If the velocity of the electron in Bohr's first orbit is 2.19 x 106 ms-1, calculate the de Broglie wavelength associated with it.

Q 2.60
The velocity associated with a proton moving in a potential difference of 1000 V is 4.37 x 105 ms-1. If the hockey ball of mass 0.1 kg is moving with this velocity, calcualte the wavelength associated with this velocity.

Q 2.61
If the position of the electron is measured within an accuracy of + 0.002 nm, calculate the uncertainty in the momentum of the electron. Suppose the momentum of the electron is h/4pim x 0.05 nm, is there any problem in defining this value.

Q 2.62
The quantum numbers of six electrons are given below. Arrange them in order of increasing energies. If any of these combination(s) has/have the same energy lists: 1. n = 4, l = 2, ml = -2 , ms = -1/2 2. n = 3, l = 2, ml = 1 , ms = +1/2 73 structure of atom 3. n = 4, l = 1, ml = 0 , ms = +1/2 4. n = 3, l = 2, ml = -2 , ms = -1/2 5. n = 3, l = 1, ml = -1 , ms = +1/2 6. n = 4, l = 1, ml = 0 , ms = +1/2

Q 2.63
The bromine atom possesses 35 electrons. It contains 6 electrons in 2p orbital, 6 electrons in 3p orbital and 5 electron in 4p orbital. Which of these electron experiences the lowest effective nuclear charge ?

Q 2.64
Among the following pairs of orbitals which orbital will experience the larger effective nuclear charge? (i) 2s and 3s, (ii) 4d and 4f, (iii) 3d and 3p.

Q 2.65
The unpaired electrons in Al and Si are present in 3p orbital. Which electrons will experience more effective nuclear charge from the nucleus ?

Q 2.66
Indicate the number of unpaired electrons in : (a) P, (b) Si, (c) Cr, (d) Fe and (e) Kr.

Q 2.67
(a) How many subshells are associated with n = 4 ? (b) How many electrons will be present in the subshells having ms value of -1/2 for n = 4 ?

Unit 3: Classification of Elements and Periodicity in Properties

38 exercises

Q 3.1
What is the basic theme of organisation in the periodic table?

Q 3.2
Which important property did Mendeleev use to classify the elements in his periodic table and did he stick to that?

Q 3.3
What is the basic difference in approach between the Mendeleev's Periodic Law and the Modern Periodic Law?

Q 3.4
On the basis of quantum numbers, justify that the sixth period of the periodic table should have 32 elements.

Q 3.5
In terms of period and group where would you locate the element with Z =114?

Q 3.6
Write the atomic number of the element present in the third period and seventeenth group of the periodic table.

Q 3.7
Which element do you think would have been named by (i) Lawrence Berkeley Laboratory
(ii) Seaborg's group?

Q 3.8
Why do elements in the same group have similar physical and chemical properties?

Q 3.9
What does atomic radius and ionic radius really mean to you?

Q 3.10
How do atomic radius vary in a period and in a group? How do you explain the variation?

Q 3.11
What do you understand by isoelectronic species? Name a species that will be isoelectronic with each of the following atoms or ions.
(i) F^-

Q 3.12
(ii) Ar
(iii) Mg^{2+}
(iv) Rb^+ Consider the following species : N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+}
(a) What is common in them?
(b) Arrange them in the order of increasing ionic radii.

Q 3.13
Explain why cation are smaller and anions larger in radii than their parent atoms?

Q 3.14
What is the significance of the terms - 'isolated gaseous atom' and 'ground state' while defining the ionization enthalpy and electron gain enthalpy? Hint : Requirements for comparison purposes.

Q 3.15
Energy of an electron in the ground state of the hydrogen atom is -2.18×10^{-18}J. Calculate the ionization enthalpy of atomic hydrogen in terms of J mol^{-1}. Hint: Apply the idea of mole concept to derive the answer.

Q 3.16
Among the second period elements the actual ionization enthalpies are in the order Li < B < Be < C < O < N < F < Ne. Explain why (i) Be has higher delta i H than B
(ii) O has lower delta i H than N and F? Classification of Elements and Periodicity in Properties

Q 3.17
How would you explain the fact that the first ionization enthalpy of sodium is lower than that of magnesium but its second ionization enthalpy is higher than that of magnesium?

Q 3.18
What are the various factors due to which the ionization enthalpy of the main group elements tends to decrease down a group?

Q 3.19
The first ionization enthalpy values (in kJ mol-1) of group 13 elements are : B Al Ga In Tl 97 How would you explain this deviation from the general trend ?

Q 3.20
Which of the following pairs of elements would have a more negative electron gain enthalpy?
(i) O or F
(ii) F or Cl

Q 3.21
Would you expect the second electron gain enthalpy of O as positive, more negative or less negative than the first? Justify your answer.

Q 3.22
What is the basic difference between the terms electron gain enthalpy and electronegativity?

Q 3.23
How would you react to the statement that the electronegativity of N on Pauling scale is 3.0 in all the nitrogen compounds?

Q 3.24
Describe the theory associated with the radius of an atom as it
(a) gains an electron
(b) loses an electron

Q 3.25
Would you expect the first ionization enthalpies for two isotopes of the same element to be the same or different? Justify your answer.

Q 3.26
What are the major differences between metals and non-metals?

Q 3.27
Use the periodic table to answer the following questions.
(a) Identify an element with five electrons in the outer subshell.
(b) Identify an element that would tend to lose two electrons.
(c) Identify an element that would tend to gain two electrons.
(d) Identify the group having metal, non-metal, liquid as well as gas at the room temperature.

Q 3.28
The increasing order of reactivity among group 1 elements is Li < Na < K < Rb <Cs whereas that among group 17 elements is F > CI > Br > I. Explain.

Q 3.29
Write the general outer electronic configuration of s-, p-, d- and f- block elements.

Q 3.30
Assign the position of the element having outer electronic configuration
(i) ns2np4 for n=3 (ii) (n-1)d2ns2 for n=4, and (iii) (n-2) f 7 (n-1)d1ns2 for n=6, in the periodic table. 98

Q 3.31

The first (delta iH1) and the second (delta iH2) ionization enthalpies (in kJ mol-1) and the (delta egH) electron gain enthalpy (in kJ mol-1) of a few elements are given below: Elements delta H1 delta H2 delta egH I 7300 -60 II 3051 -48 III 1681 3374 -328 IV 1008 1846 -295 V 2372 5251 +48 VI 1451 -40 Which of the above elements is likely to be :

(a) the least reactive element.
(b) the most reactive metal.
(c) the most reactive non-metal.
(d) the least reactive non-metal.
(e) the metal which can form a stable binary halide of the formula MX2(X=halogen). (f)

Q 3.32

the metal which can form a predominantly stable covalent halide of the formula MX (X=halogen)? Predict the formulas of the stable binary compounds that would be formed by the combination of the following pairs of elements.

(a) Lithium and oxygen
(b) Magnesium and nitrogen
(c) Aluminium and iodine
(d) Silicon and oxygen
(e) Phosphorus and fluorine (f)

Q 3.33

Element 71 and fluorine In the modern periodic table, the period indicates the value of :

(a) atomic number
(b) atomic mass
(c) principal quantum number
(d) azimuthal quantum number.

Q 3.34

Which of the following statements related to the modern periodic table is incorrect?

(a) The p-block has 6 columns, because a maximum of 6 electrons can occupy all the orbitals in a p-shell.
(b) The d-block has 8 columns, because a maximum of 8 electrons can occupy all the orbitals in a d-subshell.
(c) Each block contains a number of columns equal to the number of electrons that can occupy that subshell.
(d) The block indicates value of azimuthal quantum number (l) for the last subshell that received electrons in building up the electronic configuration. Classification of Elements and Periodicity in Properties

Q 3.35

99 Anything that influences the valence electrons will affect the chemistry of the element. Which one of the following factors does not affect the valence shell?

(a) Valence principal quantum number (n)
(b) Nuclear charge (Z)
(c) Nuclear mass
(d) Number of core electrons.

Q 3.36

The size of isoelectronic species - F-, Ne and Na+ is affected by

(a) nuclear charge (Z)
(b) valence principal quantum number (n)
(c) electron-electron interaction in the outer orbitals
(d) none of the factors because their size is the same.

Q 3.37

Which one of the following statements is incorrect in relation to ionization enthalpy?

(a) Ionization enthalpy increases for each successive electron.
(b) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.
(c) End of valence electrons is marked by a big jump in ionization enthalpy.
(d) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

Q 3.40

Considering the elements B, Al, Mg, and K, the correct order of their metallic character is :

(a) B > Al > Mg > K

(b) Al > Mg > B > K

(c) Mg > Al > K > B

(d) K > Mg > Al > B Considering the elements B, C, N, F, and Si, the correct order of their non-metallic character is :

(a) B > C > Si > N > F

(b) Si > C > B > N > F

(c) F > N > C > B > Si

(d) F > N > C > Si > B Considering the elements F, Cl, O and N, the correct order of their chemical reactivity in terms of oxidizing property is :

(a) F > Cl > O > N

(b) F > O > Cl > N

(c) Cl > F > O > N

(d) O > F > N > Cl

Unit 4: Chemical Bonding and Molecular Structure

39 exercises

Q 4.1

Explain the formation of a chemical bond.

Q 4.2

Write Lewis dot symbols for atoms of the following elements : Mg, Na, B, O, N, Br.

Q 4.3

Write Lewis symbols for the following atoms and ions: S and S2-; Al and Al3+; H and H-

Q 4.4

Draw the Lewis structures for the following molecules and ions : 2- H2S, SiCl4, BeF2, CO3 , HCOOH

Q 4.5

Define octet rule. Write its significance and limitations.

Q 4.6

4.7 Write the favourable factors for the formation of ionic bond. Discuss the shape of the following molecules using the VSEPR model: BeCl2, BCl3, SiCl4, AsF5, H2S, PH3

Q 4.8

Although geometries of NH3 and H2O molecules are distorted tetrahedral, bond angle in water is less than that of ammonia. Discuss.

Q 4.9

How do you express the bond strength in terms of bond order ?

Q 4.10

Define the bond length.

Q 4.11

Explain the important aspects of resonance with reference to the CO3 ion.

Q 4.12

H3PO3 can be represented by structures 1 and 2 shown below. Can these two structures be taken as the canonical forms of the resonance hybrid representing H3PO3 ? If not, give reasons for the same.

Q 4.13

Write the resonance structures for SO3, NO2 and NO3 .

Q 4.14

Use Lewis symbols to show electron transfer between the following atoms to form cations and anions : (a) K and S (b) Ca and O (c) Al and N.

Q 4.15

Although both CO2 and H2O are triatomic molecules, the shape of H2O molecule is bent while that of CO2 is linear. Explain this on the basis of dipole moment.

Q 4.16

Write the significance/applications of dipole moment.

Q 4.17
Define electronegativity. How does it differ from electron gain enthalpy ?

Q 4.18
Explain with the help of suitable example polar covalent bond.

Q 4.19
Arrange the bonds in order of increasing ionic character in the molecules: LiF, K_2O, N_2, SO_2 and ClF_3.

Q 4.20
The skeletal structure of CH_3COOH as shown below is correct, but some of the bonds are shown incorrectly. Write the correct Lewis structure for acetic acid.

Q 4.21
Apart from tetrahedral geometry, another possible geometry for CH_4 is square planar with the four H atoms at the corners of the square and the C atom at its centre. Explain why CH_4 is not square planar ?

Q 4.22
Explain why BeH_2 molecule has a zero dipole moment although the Be-H bonds are polar.

Q 4.23
2- - Which out of NH_3 and NF_3 has higher dipole moment and why ?

Q 4.24
What is meant by hybridisation of atomic orbitals? Describe the shapes of sp, sp^2, sp^3 hybrid orbitals.

Q 4.25
Describe the change in hybridisation (if any) of the Al atom in the following reaction. $AlCl_3 + Cl^- \rightarrow AlCl_4^-$
Chemical Bonding And Molecular Structure

Q 4.26
Is there any change in the hybridisation of B and N atoms as a result of the following reaction?

Q 4.27
Draw diagrams showing the formation of a double bond and a triple bond between carbon atoms in C_2H_4 and C_2H_2 molecules.

Q 4.28
What is the total number of sigma and pi bonds in the following molecules?
(a) C_2H_2 (b) C_2H_4

Q 4.29
Considering x-axis as the internuclear axis which out of the following will not form a sigma bond and why? (a) 1s and 1s (b) 1s and 2px; (c) 2py and 2py
(d) 1s and 2s.

Q 4.30
Which hybrid orbitals are used by carbon atoms in the following molecules?

Q 4.31
What do you understand by bond pairs and lone pairs of electrons? Illustrate by giving one exmaple of each type.
CH_3-CH_3; (b) CH_3-CH=CH_2; (c) CH_3-CH_2-OH; (d) CH_3-CHO (e) CH_3COOH

Q 4.32
Distinguish between a sigma and a pi bond.

Q 4.33
Explain the formation of H_2 molecule on the basis of valence bond theory.

Q 4.34
Write the important conditions required for the linear combination of atomic orbitals to form molecular orbitals.

Q 4.35
Use molecular orbital theory to explain why the Be_2 molecule does not exist.

Q 4.36
Compare the relative stability of the following species and indicate their magnetic properties; 2- (superoxide), O_2 (peroxide)

Q 4.37
Write the significance of a plus and a minus sign shown in representing the orbitals.

Q 4.38
Describe the hybridisation in case of PCl_5. Why are the axial bonds longer as compared to equatorial bonds?

Q 4.39
Define hydrogen bond. Is it weaker or stronger than the van der Waals forces?

Q 4.40
What is meant by the term bond order? Calculate the bond order of : N_2, O_2, O_2^+ and O_2^-.

Unit 5: Thermodynamics

18 exercises

Q 5.5
Choose the correct answer. A thermodynamic state function is a quantity (i) used to determine heat changes (ii) whose value is independent of path (iii) used to determine pressure volume work (iv) whose value depends on temperature only. For the process to occur under adiabatic conditions, the correct condition is: (i) delta T = 0 (ii) delta p = 0 (iii) q=0 (iv) w=0 The enthalpies of all elements in their standard states are: (i) unity (ii) zero (iii) <0 (iv) different for each element delta U deg of combustion of methane is - X kJ mol-1. The value of delta Hdeg is (i) = delta U deg (ii) > delta U deg (iii) < delta U deg (iv) =0 The enthalpy of combustion of methane, graphite and dihydrogen at 298 K are, -890.3 kJ mol-1 -393.5 kJ mol-1, and -285.8 kJ mol-1 respectively. Enthalpy of formation of $CH_4(g)$ will be (i) -74.8 kJ mol-1
(ii) -52.27 kJ mol-1 (iii)

Q 5.6
+74.8 kJ mol-1
(iv) +52.26 kJ mol-1. A reaction, A + B -> C + D + q is found to have a positive entropy change. The reaction will be (i) possible at high temperature (ii) possible only at low temperature (iii) not possible at any temperature (v) possible at any temperature

Q 5.7
In a process, 701 J of heat is absorbed by a system and 394 J of work is done by the system. What is the change in internal energy for the process?

Q 5.8
The reaction of cyanamide, NH_2CN (s), with dioxygen was carried out in a bomb calorimeter, and delta U was found to be -742.7 kJ mol-1 at 298 K. Calculate enthalpy change for the reaction at 298 K. $NH_2CN(g)$ + 3 O (g) -> $N_2(g)$ + $CO_2(g)$ + $H_2O(l)$ 2 2

Q 5.9
Calculate the number of kJ of heat necessary to raise the temperature of 60.0 g of aluminium from 35degC to 55degC. Molar heat capacity of Al is 24 J mol-1 K-1.

Q 5.10
Calculate the enthalpy change on freezing of 1.0 mol of water at10.0degC to ice at -10.0degC. delta fusH = 6.03 kJ mol-1 at 0degC. Cp [$H_2O(l)$] = 75.3 J mol-1 K-1 Cp [$H_2O(s)$] = 36.8 J mol-1 K-1

Q 5.11
Enthalpy of combustion of carbon to CO_2 is -393.5 kJ mol-1. Calculate the heat released upon formation of 35.2 g of CO_2 from carbon and dioxygen gas.

Q 5.12
Enthalpies of formation of CO(g), $CO_2(g)$, $N_2O(g)$ and $N_2O_4(g)$ are -110, - 393, 81 and 9.7 kJ mol-1 respectively. Find the value of delta rH for the reaction: $N_2O_4(g)$ + 3CO(g) -> $N_2O(g)$ + $3CO_2(g)$

Q 5.13
Given $N_2(g)$ + $3H_2(g)$ -> $2NH_3(g)$; delta rHdeg = -92.4 kJ mol-1 What is the standard enthalpy of formation of NH_3 gas?

Q 5.14
Calculate the standard enthalpy of formation of $CH_3OH(l)$ from the following data: 3 O (g) -> $CO_2(g)$ + $2H_2O(l)$; delta rHdeg = -726 kJ mol-1 2 2 C(graphite) + $O_2(g)$ -> $CO_2(g)$; delta cHdeg = -393 kJ mol-1 CH_3OH (l) + 1 O (g) -> $H_2O(l)$; delta f Hdeg = -286 kJ mol-1. 2 2 Calculate the enthalpy change for the process $CCl_4(g)$ -> C(g) + 4 Cl(g) and calculate bond enthalpy of C - Cl in $CCl_4(g)$. delta vapHdeg(CCl_4) = 30.5 kJ mol-1. $H_2(g)$ +

Q 5.15
delta fHdeg (CCl_4) = -135.5 kJ mol-1. delta aHdeg (C) = 715.0 kJ mol-1, where delta aHdeg is enthalpy of atomisation delta aHdeg (Cl_2) = 242 kJ mol-1

Q 5.16
For an isolated system, delta U = 0, what will be delta S ?

Q 5.17
For the reaction at 298 K, 2A + B -> C delta H = 400 kJ mol-1 and delta S = 0.2 kJ K-1 mol-1 At what temperature will the reaction become spontaneous considering delta H and delta S to be constant over the temperature range.

Q 5.18
For the reaction, 2 Cl(g) -> Cl2(g), what are the signs of delta H and delta S ?

Q 5.19
For the reaction 2 A(g) + B(g) -> 2D(g) delta U deg = -10.5 kJ and delta Sdeg = -44.1 JK-1. Calculate delta Gdeg for the reaction, and predict whether the reaction may occur spontaneously. THERMODYNAMICS

Q 5.20
The equilibrium constant for a reaction is 10. What will be the value of delta Gdeg ? R = 8.314 JK-1 mol-1, T = 300 K.

Q 5.21
Comment on the thermodynamic stability of NO(g), given 1 1 N2(g) + O (g) -> NO(g); delta rHdeg = 90 kJ mol-1 2 2 2 1 NO(g) + O2(g) -> NO2(g): delta rHdeg= -74 kJ mol-1 2

Q 5.22
Calculate the entropy change in surroundings when 1.00 mol of H2O(l) is formed under standard conditions. delta f Hdeg = -286 kJ mol-1.

Unit 6: Equilibrium

72 exercises

Q 6.1
A liquid is in equilibrium with its vapour in a sealed container at a fixed temperature. The volume of the container is suddenly increased. a) What is the initial effect of the change on vapour pressure? b) How do rates of evaporation and condensation change initially? c) What happens when equilibrium is restored finally and what will be the final vapour pressure?

Q 6.2
What is Kc for the following equilibrium when the equilibrium concentration of each substance is: [SO2]= 0.60M, [O2] = 0.82M and [SO3] = 1.90M ?

Q 6.3
At a certain temperature and total pressure of 105Pa, iodine vapour contains 40% by volume of I atoms 2SO2(g) + O2(g) I2 (g) 2SO3(g) 2I (g) Calculate Kp for the equilibrium.

Q 6.4
Write the expression for the equilibrium constant, Kc for each of the following reactions: (i) 2NOCl (g)
(ii) 2Cu(NO3)2 (s) 2NO (g) + Cl2 (g) 2CuO (s) + 4NO2 (g) + O2 (g)
(iii) CH3COOC2H5(aq) + H2O(l)
(iv) Fe3+ (aq) + 3OH- (aq)
(v) I2 (s) + 5F2

Q 6.5
CH3COOH (aq) + C2H5OH (aq) Fe(OH)3 (s) 2IF5 Find out the value of Kc for each of the following equilibria from the value of Kp: (i) 2NOCl (g)
(ii) CaCO3 (s) 2NO (g) + Cl2 (g); Kp= 1.8 x 10-2 at 500 K CaO(s) + CO2(g); Kp= 167 at 1073 K

Q 6.6
For the following equilibrium, Kc= 6.3 x 1014 at 1000 K NO (g) + O3 (g) NO2 (g) + O2 (g) Both the forward and reverse reactions in the equilibrium are elementary bimolecular reactions. What is Kc, for the reverse reaction?

Q 6.7
Explain why pure liquids and solids can be ignored while writing the equilibrium constant expression?

Q 6.8
Reaction between N2 and O2- takes place as follows: 2N2 (g) + O2 (g) 2N2O (g) If a mixture of 0.482 mol N2 and 0.933 mol of O2 is placed in a 10 L reaction vessel and allowed to form N2O at a temperature for which Kc= 2.0 x 10-37, determine the composition of equilibrium mixture.

Q 6.9

Nitric oxide reacts with Br_2 and gives nitrosyl bromide as per reaction given below: 2NO (g) + Br_2 (g) 2NOBr (g) When 0.087 mol of NO and 0.0437 mol of Br_2 are mixed in a closed container at constant temperature, 0.0518 mol of NOBr is obtained at equilibrium. Calculate equilibrium amount of NO and Br_2 .

Q 6.10

At 450K, Kp= 2.0 x 1010/bar for the given reaction at equilibrium. $2SO_2$(g) + O_2(g) $2SO_3$ (g) What is Kc at this temperature ?

Q 6.11

A sample of HI(g) is placed in flask at a pressure of 0.2 atm. At equilibrium the partial pressure of HI(g) is 0.04 atm. What is Kp for the given equilibrium ? 2HI (g) H_2 (g) + I_2 (g)

Q 6.12

A mixture of 1.57 mol of N_2, 1.92 mol of H_2 and 8.13 mol of NH_3 is introduced into a 20 L reaction vessel at 500 K. At this temperature, the equilibrium constant, Kc for the reaction N_2 (g) + $3H_2$ (g) $2NH_3$ (g) is 1.7 x 102. Is the reaction mixture at equilibrium? If not, what is the direction of the net reaction?

Q 6.13

The equilibrium constant expression for a gas reaction is, [NH_3] [O_2]5 6 [NO]4 [H_2O] 4 Kc = Write the balanced chemical equation corresponding to this expression.

Q 6.14

One mole of H_2O and one mole of CO are taken in 10 L vessel and heated to 725 K. At equilibrium 40% of water (by mass) reacts with CO according to the equation, H_2O (g) + CO (g) H_2 (g) + CO_2 (g) Calculate the equilibrium constant for the reaction.

Q 6.15

At 700 K, equilibrium constant for the reaction: H_2 (g) + I_2 (g) 2HI (g) is 54.8. If 0.5 mol L of HI(g) is present at equilibrium at 700 K, what are the concentration of H_2(g) and I_2(g) assuming that we initially started with HI(g) and allowed it to reach equilibrium at 700K? -1

Q 6.16

What is the equilibrium concentration of each of the substances in the equilibrium when the initial concentration of ICl was 0.78 M ? 2ICl (g)

Q 6.17

I_2 (g) + Cl_2 (g); Kc = 0.14 Kp = 0.04 atm at 899 K for the equilibrium shown below. What is the equilibrium concentration of C_2H_6 when it is placed in a flask at 4.0 atm pressure and allowed to come to equilibrium?

Q 6.18

C_2H_6 (g) C_2H_4 (g) + H_2 (g) Ethyl acetate is formed by the reaction between ethanol and acetic acid and the equilibrium is represented as: CH_3COOH (l) + C_2H_5OH (l) (i) $CH_3COOC_2H_5$ (l) + H_2O (l) Write the concentration ratio (reaction quotient), Qc, for this reaction (note: water is not in excess and is not a solvent in this reaction)

(ii) At 293 K, if one starts with 1.00 mol of acetic acid and 0.18 mol of ethanol, there is 0.171 mol of ethyl acetate in the final equilibrium mixture. Calculate the equilibrium constant.

(iii) Starting with 0.5 mol of ethanol and 1.0 mol of acetic acid and maintaining it at 293 K, 0.214 mol of ethyl acetate is found after sometime. Has equilibrium been reached?

Q 6.19

A sample of pure PCl 5 was introduced into an evacuated vessel at 473 K. After equilibrium was attained, concentration of PCl 5 was found to be 0.5 x 10-1 mol L-1. If value of Kc is 8.3 x 10-3, what are the concentrations of PCl_3 and Cl_2 at equilibrium? PCl_5 (g)

Q 6.20

PCl_3 (g) + Cl_2(g) One of the reaction that takes place in producing steel from iron ore is the reduction of iron(II) oxide by carbon monoxide to give iron metal and CO_2. FeO (s) + CO (g) Fe (s) + CO_2 (g); Kp = 0.265 atm at 1050K What are the equilibrium partial pressures of CO and CO_2 at 1050 K if the initial = 0.80 atm? partial pressures are: pCO= 1.4 atm and

Q 6.21

Equilibrium constant, Kc for the reaction N_2 (g) + $3H_2$ (g) $2NH_3$ (g) at 500 K is 0.061 At a particular time, the analysis shows that composition of the reaction mixture is 3.0 mol L-1 N_2, 2.0 mol L-1 H_2 and 0.5 mol L-1 NH_3. Is the reaction at equilibrium? If not in which direction does the reaction tend to proceed to reach equilibrium?

Q 6.22

Bromine monochloride, BrCl decomposes into bromine and chlorine and reaches the equilibrium: 2BrCl (g) Br2 (g) + Cl2 (g) for which Kc= 32 at 500 K. If initially pure BrCl is present at a concentration of 3.3 x 10-3 mol L-1, what is its molar concentration in the mixture at equilibrium?

Q 6.23

At 1127 K and 1 atm pressure, a gaseous mixture of CO and CO2 in equilibrium with soild carbon has 90.55% CO by mass C (s) + CO2 (g) 2CO (g) Calculate Kc for this reaction at the above temperature.

Q 6.24

6.25 Calculate a) delta Gdeg and b) the equilibrium constant for the formation of NO2 from NO and O2 at 298K NO (g) + 1/2 O2 (g) NO2 (g) where delta fGdeg (NO2) = 52.0 kJ/mol delta fGdeg (NO) = 87.0 kJ/mol delta fGdeg (O2) = 0 kJ/mol Does the number of moles of reaction products increase, decrease or remain same when each of the following equilibria is subjected to a decrease in pressure by increasing the volume?

(a) PCl5 (g)

Q 6.26

PCl3 (g) + Cl2 (g)

(b) CaO (s) + CO2 (g) CaCO3 (s)

(c) 3Fe (s) + 4H2O (g) Fe3O4 (s) + 4H2 (g) Which of the following reactions will get affected by increasing the pressure? Also, mention whether change will cause the reaction to go into forward or backward direction. (i) COCl2 (g) CO (g) + Cl2 (g) CS2 (g) + 2H2S (g)

(ii) CH4 (g) + 2S2 (g) 2CO (g)

(iii) CO2 (g) + C (s) CH3OH (g)

(iv) 2H2 (g) + CO (g)

(v) CaCO3 (s) CaO (s) + CO2 (g)

(vi) 4 NH3 (g) + 5O2 (g)

Q 6.27

4NO (g) + 6H2O(g) The equilibrium constant for the following reaction is 1.6 x105 at 1024K H2(g) + Br2(g) 2HBr(g) Find the equilibrium pressure of all gases if 10.0 bar of HBr is introduced into a sealed container at 1024K.

Q 6.28

Dihydrogen gas is obtained from natural gas by partial oxidation with steam as per following endothermic reaction: CH4 (g) + H2O (g) CO (g) + 3H2 (g)

(a) Write as expression for Kp for the above reaction.

(b) How will the values of Kp and composition of equilibrium mixture be affected by (i) increasing the pressure (ii) increasing the temperature

(iii) using a catalyst?

Q 6.29

Describe the effect of: a) addition of H2 b) addition of CH3OH c) removal of CO d) removal of CH3OH on the equilibrium of the reaction: 2H2(g) + CO (g)

Q 6.30

CH3OH (g) At 473 K, equilibrium constant Kc for decomposition of phosphorus pentachloride, PCl5 is 8.3 x10-3. If decomposition is depicted as, PCl5 (g)

Q 6.31

delta rH deg = 124.0 kJ mol-1 PCl3 (g) + Cl2 (g) a) write an expression for Kc for the reaction. b) what is the value of Kc for the reverse reaction at the same temperature? c) what would be the effect on Kc if (i) more PCl5 is added

(ii) pressure is increased

(iii) the temperature is increased ? Dihydrogen gas used in Haber's process is produced by reacting methane from natural gas with high temperature steam. The first stage of two stage reaction involves the formation of CO and H2. In second stage, CO formed in first stage is reacted with more steam in water gas shift reaction, CO (g) + H2O (g) CO2 (g) + H2 (g) If a reaction vessel at 400 degC is charged with an equimolar mixture of CO and steam such that pco = pH O = 4.0 bar, what will be the partial pressure of H2 at equilibrium? 2 Kp= 10.1 at 400degC

Q 6.32

Predict which of the following reaction will have appreciable concentration of reactants and products: a) Cl2 (g) b) Cl2 (g) + 2NO (g) 2NOCl (g) c) Cl2 (g) + 2NO2 (g) 2NO2Cl (g) Kc = 1.8 2Cl (g) Kc = 5 x10-39 Kc = 3.7 x 108

Q 6.33

2O3 (g) is 2.0 x10-50 at 25degC. If the The value of Kc for the reaction 3O2 (g) equilibrium concentration of O2 in air at 25degC is 1.6 x10-2, what is the concentration of O3?

Q 6.34
The reaction, $CO(g) + 3H_2(g)$ $CH_4(g) + H_2O(g)$ is at equilibrium at 1300 K in a 1L flask. It also contain 0.30 mol of CO, 0.10 mol of H_2 and 0.02 mol of H_2O and an unknown amount of CH_4 in the flask. Determine the concentration of CH_4 in the mixture. The equilibrium constant, Kc for the reaction at the given temperature is 3.90.

Q 6.35
What is meant by the conjugate acid-base pair? Find the conjugate acid/base for the following species: HNO_2, CN^-, $HClO_4$, F^-, OH^-, CO , and S^{2-}

Q 6.36
Which of the followings are Lewis acids? H_2O, BF_3, H^+, and NH_4^+

Q 6.37
What will be the conjugate bases for the Bronsted acids: HF, H_2SO_4 and HCO_3^-?

Q 6.38
Write the conjugate acids for the following Bronsted bases: NH_2^-, NH_3 and $HCOO^-$.

Q 6.39
The species: H_2O, HCO_3^-, HSO_4^- and NH_3 can act both as Bronsted acids and bases. For each case give the corresponding conjugate acid and base.

Q 6.40
Classify the following species into Lewis acids and Lewis bases and show how these act as Lewis acid/base: (a) OH^- (b) F^- (c) H^+ (d) BCl_3 .

Q 6.41
The concentration of hydrogen ion in a sample of soft drink is 3.8×10^{-3} M. What is its pH?

Q 6.42
The pH of a sample of vinegar is 3.76. Calculate the concentration of hydrogen ion in it.

Q 6.43
The ionization constant of HF, HCOOH and HCN at 298K are 6.8×10^{-4}, 1.8×10^{-4} and 4.8×10^{-9} respectively. Calculate the ionization constants of the corresponding conjugate base.

Q 6.44
The ionization constant of phenol is 1.0×10^{-10}. What is the concentration of phenolate ion in 0.05 M solution of phenol? What will be its degree of ionization if the solution is also 0.01M in sodium phenolate?

Q 6.45
The first ionization constant of H_2S is 9.1×10^{-8}. Calculate the concentration of HS^- ion in its 0.1M solution. How will this concentration be affected if the solution is 0.1M in HCl also? If the second dissociation constant of H_2S is 1.2×10^{-13}, calculate the concentration of S^{2-} under both conditions.

Q 6.46
The ionization constant of acetic acid is 1.74×10^{-5}. Calculate the degree of dissociation of acetic acid in its 0.05 M solution. Calculate the concentration of acetate ion in the solution and its pH.

Q 6.47
It has been found that the pH of a 0.01M solution of an organic acid is 4.15. Calculate the concentration of the anion, the ionization constant of the acid and its pKa.

Q 6.48
Assuming complete dissociation, calculate the pH of the following solutions:
(a) 0.003 M HCl (b) 0.005 M NaOH (c) 0.002 M HBr (d) 0.002 M KOH

Q 6.49
Calculate the pH of the following solutions: a) 2 g of TlOH dissolved in water to give 2 litre of solution. b) 0.3 g of $Ca(OH)_2$ dissolved in water to give 500 mL of solution. c) 0.3 g of NaOH dissolved in water to give 200 mL of solution. d) 1mL of 13.6 M HCl is diluted with water to give 1 litre of solution.

Q 6.50
The degree of ionization of a 0.1M bromoacetic acid solution is 0.132. Calculate the pH of the solution and the pKa of bromoacetic acid.

Q 6.51
The pH of 0.005M codeine ($C_{18}H_{21}NO_3$) solution is 9.95. Calculate its ionization constant and pKb.

Q 6.52
What is the pH of 0.001M aniline solution? The ionization constant of aniline can be taken from Table 6.7. Calculate the degree of ionization of aniline in the solution. Also calculate the ionization constant of the conjugate acid of aniline.

Q 6.53
Calculate the degree of ionization of 0.05M acetic acid if its pKa value is 4.74. How is the degree of dissociation affected when its solution also contains
(a) 0.01M (b) 0.1M in HCl ?

Q 6.54
The ionization constant of dimethylamine is 5.4 x 10-4. Calculate its degree of ionization in its 0.02M solution. What percentage of dimethylamine is ionized if the solution is also 0.1M in NaOH?

Q 6.55
Calculate the hydrogen ion concentration in the following biological fluids whose pH are given below:
(a) Human muscle-fluid, 6.83
(b) Human stomach fluid, 1.2
(c) Human blood, 7.38
(d) Human saliva, 6.4.

Q 6.56
The pH of milk, black coffee, tomato juice, lemon juice and egg white are 6.8, 5.0, 4.2, 2.2 and 7.8 respectively. Calculate corresponding hydrogen ion concentration in each.

Q 6.57
If 0.561 g of KOH is dissolved in water to give 200 mL of solution at 298 K. Calculate the concentrations of potassium, hydrogen and hydroxyl ions. What is its pH?

Q 6.58
The solubility of Sr(OH)2 at 298 K is 19.23 g/L of solution. Calculate the concentrations of strontium and hydroxyl ions and the pH of the solution.

Q 6.59
The ionization constant of propanoic acid is 1.32 x 10-5. Calculate the degree of ionization of the acid in its 0.05M solution and also its pH. What will be its degree of ionization if the solution is 0.01M in HCl also?

Q 6.60
The pH of 0.1M solution of cyanic acid (HCNO) is 2.34. Calculate the ionization constant of the acid and its degree of ionization in the solution.

Q 6.61
The ionization constant of nitrous acid is 4.5 x 10-4. Calculate the pH of 0.04 M sodium nitrite solution and also its degree of hydrolysis.

Q 6.62
A 0.02M solution of pyridinium hydrochloride has pH = 3.44. Calculate the ionization constant of pyridine.

Q 6.63
Predict if the solutions of the following salts are neutral, acidic or basic: NaCl, KBr, NaCN, NH4NO3, NaNO2 and KF

Q 6.64
The ionization constant of chloroacetic acid is 1.35 x 10-3. What will be the pH of 0.1M acid and its 0.1M sodium salt solution?

Q 6.65
Ionic product of water at 310 K is 2.7 x 10-14. What is the pH of neutral water at this temperature?

Q 6.66
Calculate the pH of the resultant mixtures: a) 10 mL of 0.2M Ca(OH)2 + 25 mL of 0.1M HCl b) 10 mL of 0.01M H2SO4 + 10 mL of 0.01M Ca(OH)2 c) 10 mL of 0.1M H2SO4 + 10 mL of 0.1M KOH

Q 6.67
Determine the solubilities of silver chromate, barium chromate, ferric hydroxide, lead chloride and mercurous iodide at 298K from their solubility product constants given in Table 6.9. Determine also the molarities of individual ions.

Q 6.68
The solubility product constant of Ag 2CrO 4 and AgBr are 1.1 x 10 -12 and 5.0 x 10-13 respectively. Calculate the ratio of the molarities of their saturated solutions.

Q 6.69

Equal volumes of 0.002 M solutions of sodium iodate and cupric chlorate are mixed together. Will it lead to precipitation of copper iodate? (For cupric iodate Ksp = 7.4 x 10-8).

Q 6.70

The ionization constant of benzoic acid is 6.46 x 10-5 and Ksp for silver benzoate is 2.5 x 10-13. How many times is silver benzoate more soluble in a buffer of pH 3.19 compared to its solubility in pure water?

Q 6.71

What is the maximum concentration of equimolar solutions of ferrous sulphate and sodium sulphide so that when mixed in equal volumes, there is no precipitation of iron sulphide? (For iron sulphide, Ksp = 6.3 x 10-18).

Q 6.72

What is the minimum volume of water required to dissolve 1g of calcium sulphate at 298 K? (For calcium sulphate, Ksp is 9.1 x 10-6).

Q 6.73

The concentration of sulphide ion in 0.1M HCl solution saturated with hydrogen sulphide is 1.0 x 10-19 M. If 10 mL of this is added to 5 mL of 0.04 M solution of the following: FeSO4, MnCl2, ZnCl2 and CdCl2. in which of these solutions precipitation will take place?

Unit 7: Redox Reactions

25 exercises

Q 7.1

7.2 Assign oxidation number to the underlined elements in each of the following species:

(a) NaH2PO4

(b) NaHSO4

(c) H4P2O7

(d) K2MnO4

(e) CaO2 (f) NaBH4 (g) H2S2O7 (h) KAl(SO4)2.12 H2O What are the oxidation number of the underlined elements in each of the following and how do you rationalise your results ?

(a) KI3

Q 7.3

(b) H2S4O6

(c) Fe3O4

(d) CH3CH2OH

(e) CH3COOH Justify that the following reactions are redox reactions: (a) CuO(s) + H2(g) -> Cu(s) + H2O(g)

(b) Fe2O3(s) + 3CO(g) -> 2Fe(s) + 3CO2(g) (c) 4BCl3(g) + 3LiAlH4(s) -> 2B2H6(g) + 3LiCl(s) + 3 AlCl3 (s)

(d) 2K(s) + F2(g) -> 2K+F- (s) (e)

Q 7.4

4 NH3(g) + 5 O2(g) -> 4NO(g) + 6H2O(g) Fluorine reacts with ice and results in the change: H2O(s) + F2(g) -> HF(g) + HOF(g) Justify that this reaction is a redox reaction.

Q 7.5

7.6 Calculate the oxidation number of sulphur, chromium and nitrogen in H2SO5, Cr2O72- and NO3-. Suggest structure of these compounds. Count for the fallacy. Write formulas for the following compounds: (a) Mercury(II) chloride

(b) Nickel(II) sulphate (c) Tin(IV) oxide

(d) Thallium(I) sulphate (e) Iron(III) sulphate (f) Chromium(III) oxide 77 ggest a list o t e s to + and nitrogen ro stances -3 to +5 ere car on can ex i it oxidation states ro -

Q 7.8

While sulphur dioxide and hydrogen peroxide can act as oxidising as well as reducing agents in their reactions, ozone and nitric acid act only as oxidants. Why ?

Q 7.9

Consider the reactions: (a) 6 CO2(g) + 6H2O(l) -> C6 H12 O6(aq) + 6O2(g) REDOX REACTIONS

(b) O3(g) + H2O2(l) -> H2O(l) + 2O2(g) Why it is more appropriate to write these reactions as : (a) 6CO2(g) + 12H2O(l) -> C6 H12 O6(aq) + 6H2O(l) + 6O2(g)

(b) O3(g) + H2O2 (l) -> H2O(l) + O2(g) + O2(g) Also suggest a technique to investigate the path of the above (a) and (b) redox reactions.

Q 7.10

The compound AgF_2 is unstable compound. However, if formed, the compound acts as a very strong oxidising agent. Why ?

Q 7.11

Whenever a reaction between an oxidising agent and a reducing agent is carried out, a compound of lower oxidation state is formed if the reducing agent is in excess and a compound of higher oxidation state is formed if the oxidising agent is in excess. Justify this statement giving three illustrations.

Q 7.12

How do you count for the following observations ?

(a) Though alkaline potassium permanganate and acidic potassium permanganate both are used as oxidants, yet in the manufacture of benzoic acid from toluene we use alcoholic potassium permanganate as an oxidant. Why ? Write a balanced redox equation for the reaction.

(b) When concentrated sulphuric acid is added to an inorganic mixture containing chloride, we get colourless pungent smelling gas HCl, but if the mixture contains bromide then we get red vapour of bromine. Why ?

Q 7.13

Identify the substance oxidised reduced, oxidising agent and reducing agent for each of the following reactions: (a) $2AgBr (s) + C_6H_6O_2(aq) \rightarrow 2Ag(s) + 2HBr (aq) + C_6H_4O_2(aq)$

(b) $HCHO(l) + 2[Ag (NH_3)_2]^+(aq) + 3OH^-(aq) \rightarrow 2Ag(s) + HCOO^-(aq) + 4NH_3(aq) + 2H_2O(l)$ (c) HCHO (l) + 2 Cu (aq) + 5 OH (aq) -> $Cu_2O(s)$ + HCOO (aq) + $3H_2O(l)$ 2+ - - (d) $N_2H_4(l) + 2H_2O_2(l) \rightarrow N_2(g) + 4H_2O(l)$ (e)

Q 7.14

$Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$ Consider the reactions : 2 S2O32- (aq) + I2(s) -> S4 O62-(aq) + 2I-(aq) S2O32-(aq) + 2Br2(l) + 5 H2O(l) -> 2SO42-(aq) + 4Br-(aq) + 10H+(aq) Why does the same reductant, thiosulphate react differently with iodine and bromine ?

Q 7.15

Justify giving reactions that among halogens, fluorine is the best oxidant and among hydrohalic compounds, hydroiodic acid is the best reductant.

Q 7.16

Why does the following reaction occur ? XeO6 - (aq) + 2F (aq) + 6H+(aq) -> XeO3(g)+ F2(g) + 3H2O(l) - What conclusion about the compound Na_4XeO_6 (of which XeO6- is a part) can be drawn from the reaction.

Q 7.17

Consider the reactions: (a) $H_3PO_2(aq) + 4 AgNO_3(aq) + 2 H_2O(l) \rightarrow H_3PO_4(aq) + 4Ag(s) + 4HNO_3(aq)$

(b) $H_3PO_2(aq) + 2CuSO_4(aq) + 2 H_2O(l) \rightarrow H_3PO_4(aq) + 2Cu(s) + H_2SO_4(aq)$ (c) $C_6H_5CHO(l) + 2[Ag(NH_3)_2]^+(aq) + 3OH^-(aq) \rightarrow C_6H_5COO^-(aq) + 2Ag(s) + 4NH_3 (aq) + 2 H_2O(l)$

(d) $C_6H_5CHO(l)$ + 2Cu (aq) + 5OH (aq) -> No change observed. 2+ - What inference do you draw about the behaviour of Ag^+ and Cu^{2+} from these reactions ? 71 alance t e ollo ing redox reactions ion - electron et od

(a) $MnO_4^- (aq) + I^- (aq) \rightarrow MnO_2 (s) + I_2(s)$ (in basic medium)

(b) $MnO_4^- (aq) + SO_2 (g) \rightarrow Mn^{2+} (aq) + HSO_4^- (aq)$ (in acidic solution) (c) $H_2O_2 (aq) + Fe^{2+} (aq) \rightarrow Fe^{3+} (aq) + H_2O (l)$ (in acidic solution)

(d) $Cr_2O_7^{2-} + SO_2(g) \rightarrow Cr^{3+} (aq) + SO_4^{2-} (aq)$ (in acidic solution)

Q 7.19

Balance the following equations in basic medium by ion-electron method and oxidation number methods and identify the oxidising agent and the reducing agent.

(a) $P_4(s) + OH^-(aq) \rightarrow PH_3(g) + HPO_2^- (aq)$

(b) $N_2H_4(l) + ClO_3^-(aq) \rightarrow NO(g) + Cl^-(g)$

(c) $Cl_2O_7 (g) + H_2O_2(aq) \rightarrow ClO_2^-(aq) + O_2(g) + H^+$

Q 7.20

What sorts of informations can you draw from the following reaction ? $(CN)_2(g) + 2OH^-(aq) \rightarrow CN^-(aq) + CNO^-(aq) + H_2O(l)$

Q 7.21

The Mn^{3+} ion is unstable in solution and undergoes disproportionation to give Mn^{2+}, MnO_2, and H^+ ion. Write a balanced ionic equation for the reaction.

Q 7.22

Consider the elements : Cs, Ne, I and F

(a) Identify the element that exhibits only negative oxidation state.

(b) Identify the element that exhibits only postive oxidation state.

(c) Identify the element that exhibits both positive and negative oxidation states.

Q 7.23

(d) Identify the element which exhibits neither the negative nor does the positive oxidation state. Chlorine is used to purify drinking water. Excess of chlorine is harmful. The excess of chlorine is removed by treating with sulphur dioxide. Present a balanced equation for this redox change taking place in water.

Q 7.24

Refer to the periodic table given in your book and now answer the following questions:
(a) Select the possible non metals that can show disproportionation reaction.
(b) Select three metals that can show disproportionation reaction.

Q 7.25

In Ostwald's process for the manufacture of nitric acid, the first step involves the oxidation of ammonia gas by oxygen gas to give nitric oxide gas and steam. What is the maximum weight of nitric oxide that can be obtained starting only with 10.00 g. of ammonia and 20.00 g of oxygen ?

Q 7.26

Using the standard electrode potentials given in the Table 8.1, predict if the reaction between the following is feasible:
(a) Fe3+(aq) and I-(aq)
(b) Ag+(aq) and Cu(s)
(c) Fe3+ (aq) and Cu(s)
(d) Ag(s) and Fe3+(aq)
(e) Br2(aq) and Fe2+(aq). REDOX REACTIONS

Q 7.27

7.28 Predict the products of electrolysis in each of the following: (i) An aqueous solution of AgNO3 with silver electrodes (ii) An aqueous solution AgNO3 with platinum electrodes (iii) A dilute solution of H2SO4 with platinum electrodes (iv) An aqueous solution of CuCl2 with platinum electrodes. Arrange the following metals in the order in which they displace each other from the solution of their salts. Al, Cu, Fe, Mg and Zn.

Q 7.29

Given the standard electrode potentials, K+ -2 3 , g+/Ag = 0.80V, Hg2+/Hg = 0.79V Mg2+ g -2 37 Cr3+ Cr -0 7 arrange these metals in their increasing order of reducing power.

Q 7.30

Depict the galvanic cell in which the reaction Zn(s) + 2Ag+(aq) -> Zn2+(aq) +2Ag(s) takes place, Further show: (i) which of the electrode is negatively charged, (ii) the carriers of the current in the cell, and (iii) individual reaction at each electrode.

Unit 8: Organic Chemistry: Basic Principles and Techniques

23 exercises

Q 8.1

ch ch ch sp sp h sp ch c chch hc cch chch solution h t p p sp s c-c c-h c c c-c c-h c c c c

Q 8.2

t sp sp s sp sp ch c t hconh t s t s co ch cn ch ch chcn solution sp sp sp ch sp sp sp sp sp sp sp sp sp

Q 8.3

sp t hc o 8.2.2 some characteristic Features of Bonds hc n solution sp sp i p ch sp structural rePresentatiOns OF Organic cOmPOunds 8.3.1 complete, condensed and Bond-line structural Formulas s t s s ch ch ch ch ch ch ch ch ch ch ch s 8.3 i s t t l t - s -ch a l s p m ch t ch ch s ch oh ch ch chch ch ch ch ch s s ch t t s ch ch t s h c ch hc ch ch chb ch ch ch oh organic chemistry - some basic principles and techniques i solution c c ho ch ch ch ch ch hoch cn b c

Q 8.4

e e ch ch coch ch ch ch ch ch ch solution solution

Q 8.5

hoch ch ch ch ch ch ch ch molecular models m t c s p i t i s b c 8.3.2 t three-dimensional representation of Organic molecules d c c t sp t sp b i i d i t t s d p Fig. 8.1 s Fig. 8.2 p s organic chemistry - some basic principles and techniques 8.4 classiFicatiOn OF Organic cOmPOunds t c o c c s t t t

(b) aromatic compounds a y u t l s s i. Acyclic or open chain compounds aliphatic t Benzenoid aromatic compounds ch ch e i b a n Non-benzenoid compound a a ii C y c l i c o r c l o s e d c h a i n o r r i n g compounds

(a) alicyclic compounds a t Heterocyclic aromatic compounds l t s s p o s ss t b c s 8.4.1 Functional group t r b c t c -oh -cho - cooh table 8.1 8.4.2 homologous series a t common or trivial names of some Organic compounds s t -ch t s i t 8.5 nOmenclature OF Organic cOmPOunds o 8.5.1 the iuPac system of nomenclature a i iuPac (international union of Pure and applied chemistry) i b s iupac ORGANIC CHEMISTRY - SOME BASIC PRINCIPLES AND TECHNIQUES By further using prefixes and suffixes, the parent name can be modified to obtain the actual name. Compounds containing carbon and hydrogen only are called hydrocarbons. A hydrocarbon is termed saturated if it contains only carbon-carbon single bonds. The IUPAC name for a homologous series of such compounds is alkane. Paraffin (Latin: little affinity) was the earlier name given to these compounds. Unsaturated hydrocarbons are those, which contain at least one carboncarbon double or triple bond. 8.5.2 IUPAC Nomenclature of Alkanes Straight chain hydrocarbons: The names of such compounds are based on their chain structure, and end with suffix '-ane' and carry a prefix indicating the number of carbon atoms present in the chain (except from CH_4 to C_4H_{10}, where the prefixes are derived from trivial names). The IUPAC names of some straight chain saturated hydrocarbons are given in Table 8.2. The alkanes in Table 8.2 differ from each other by merely the number of $-CH_2$ groups in the chain. They are homologues of alkane series. Table 8.2 IUPAC Names of Some Unbranched Saturated Hydrocarbons In order to name such compounds, the names of alkyl groups are prefixed to the name of parent alkane. An alkyl group is derived from a saturated hydrocarbon by removing a hydrogen atom from carbon. Thus, CH_4 becomes $-CH_3$ and is called methyl group. An alkyl group is named by substituting 'yl' for 'ane' in the corresponding alkane. Some alkyl groups are listed in Table 8.3. Table 8.3 Some Alkyl Groups Abbreviations are used for some alkyl groups. For example, methyl is abbreviated as Me, ethyl as Et, propyl as Pr and butyl as Bu. The alkyl groups can be branched also. Thus, propyl and butyl groups can have branched structures as shown below. CH_3-$CHCH_3$-CH-$CHCH_3$ | CH_3 | CH_3 Isopropyl- sec-Butyl- CH_3 | CH_3-C| CH_3 tert-Butyl- Branched chain hydrocarbons: In a branched chain compound small chains of carbon atoms are attached at one or more carbon atoms of the parent chain. The small carbon chains (branches) are called alkyl groups. For example: CH_3-CH-CH_2-CH_3 CH_3-CH-CH_2-CH-CH_3 | | | CH_3 CH_2CH_3 CH_3 (a) (b) CH_3 CH-CH_2- Isobutyl- CH_3 | CH_3-C-CH_2| CH_3 Neopentyl- Common branched groups have specific trivial names. For example, the propyl groups can either be n-propyl group or isopropyl group. The branched butyl groups are called sec-butyl, isobutyl and tert-butyl group. We also encounter the structural unit, $-CH_2C(CH_3)_3$, which is called neopentyl group. Nomenclature of branched chain alkanes: We encounter a number of branched chain alkanes. The rules for naming them are given below. longest carbon chain in the molecule is identified i i p s ii i t t ch ch ch ch ch ch ch ch ch ch - c -ch -ch -ch ch t p p hchc ch ch -ch -ch-c-ch -ch -ch the numbering is done in such a way that the branched carbon atoms get the lowest possible numbers. t ch p i lower number is given to the one coming first in the alphabetical listing t c-c-c-c-c-c-c-c c ch c-c c-c ch ch ch ch ch c-c-c-c-c-c-c-c c ch ch ch ch ch t h carbon atom of the branch that attaches to the root alkane t i ch -ch-ch -ch- t ch n ch organic chemistry - some basic principles and techniques t Cyclic Compounds: a i in alphabetical order, the prefixes iso- and neo- are considered to be the part of the fundamental name of alkyl group. the prefixes sec- and tert- are not t n s iupac i i a

Q 8.7

s iupac e p p solution l s s p p 8.5.3 nomenclature of Organic compounds having Functional group(s) a p p i c ch oh ch ch oh oh ch choh - - ch oh ch oh h t - ch ch ch ch - e -

Q 8.8

iupac t the longest chain of carbon atoms containing the functional group is numbered in such a way that the functional group is attached at the carbon atom possessing lowest possible number in the chain b t solution t oh h t oh h i p p p t t oh i h t m the order of decreasing priority for some functional groups is: -cOOh, -sO3h, -cOOr (r=alkyl group), cOcl, -cOnh2, -cn,-hc=O, >c=O, -Oh, -nh2, > c=c<, -c c- . c b i -no t -r c h -or t solution t c o p t t hoch ch s ch coch t b ch ch ch h i organic chemistry - some basic principles and techniques table 8.4 some Functional groups and classes of Organic compounds

(iii) s solution h i t t c t ii d n no - cooh t -no t t o (iv) solution t oh c c c oh c c t c c ii t t t h (v)

Q 8.9

d c p t n c oh h t ch ch oh - ch ch ch ch cho c cho solution (i) t h ch ch ch ch ch c ch 8.5.4 (ii) nomenclature of substituted Benzene compounds iupac c c oh t h ch chch ch oh ch organic chemistry - some basic principles and techniques s t t a s a i i p p s t p p p t i p ch p

Q 8.10

e pn ch d e ch chch ch solution ch ch ch ch ch p ch i m ch ch - c- ch ch n d

(ii) Position isomerism: 8.6 isOmerism t c h o oh s t 8.6.1 structural isomerism c ch ch ch oh ch ch ch p p

(iii) Functional group isomerism: t s cho

(i) Chain isomerism: i s c s p m g o organic chemistry - some basic principles and techniques o h y x ch c ch ch ch p c o i p

(iv) Metamerism: i ch o ch oc h c h oc h 8.6.2 stereoisomerism t 8.7.1 Fission of a covalent Bond a heterolytic cleavage, cleavage. i heterolytic cleavage homolytic a t s 8.7 t s F u n d a m e n ta l c O n c e P t s i n Organic reactiOn mechanism ch b - i a h t c t a r o i p s ch c h ch b s ch ch s c c i a s s i t a sequential account of each step, describing details of electron movement, energetics during bond cleavage and bond formation, and the rates of transformation of reactants into products (kinetics) is referred to as reaction mechanism. t ch ch ch ch ch ch c t sp ch t c sp s s e c sp -h s t t s l a a a s Fig. 8.3 (a) p a p t m e i t o p s p 8.7.2 substrate and reagent i m i c i e Fig. 8.3 (b) p ch s c ch b r -> b ch - ch b p t ionic heteropolar i homolytic cleavage t Nucleophiles and Electrophiles t r t organic chemistry - some basic principles and techniques

Q 8.11

u i i ch -sch pi ch -cn ch -c solution a nucleophile n nucleophilic a electrophile e electrophilic

Q 8.12

g d s Cl,CH C t O, H N , N O solution p a s hs- c h o- ch nhn- t s p ho- rc- n nc- b c h -c o n o

Q 8.13

i ch ch o ch cn ch i e ch solution c o r c s r a h c -i t ch hc o h cc n i c 8.7.3 electron movement in Organic reactions t i 8.7.5 inductive effect t s b p l ch ch c c-c i - t m - ch - -> -ch - -> -c i cc c - i c-c 8.7.4 s electron displacement effects in covalent Bonds t inductive effect t t t b i t h no coor t cn oa cooh - oc h o -ch -ch -ch i organic chemistry - some basic principles and techniques

Q 8.14

h c nh h c sh hch hcb h c oh h c oh solution c-b h b c-o c-o i ii s s s the resonance structures (canonical structures or contributing structures) are hypothetical and individually do not represent any real molecule. t a

Q 8.15

i c-c ch ch ch b ii t ch no l no i solution m h 8.7.6 resonance structure t l a i c-c however, it is known that the two n-O bonds of nitromethane are of the same length (intermediate between a n-O single bond and a n=O double bond). the actual structure of nitromethane is therefore a resonance hybrid of the two canonical forms i and ii. t c c t b a c-c c c resonance stabilisation energy resonance energy t c-c r h c-c c c t t t i a ii t u

Q 8.16

ch coo- solution t a i solution 8.7.7 t t resonance effect t t r m

(i) Positive Resonance Effect (+R effect) i

Q 8.17

ch ch-cho i t solution t s i ii

(ii) Negative Resonance Effect (- R effect) t iii i m ii iii

Q 8.18

e i ii ch cooch t r -r r - -oh -or -ocor -nh -nhr -nr -nhcor -r - cooh -cho c o - cn -no organic chemistry - some basic principles and techniques t t p c h t t h p i t ch c h 8.7.8 i electromeric effect (e effect) t multiple bond o p p ch i ch p p t i e t p e e e i Fig. 8.4(a) s p t n 8.7.9 h e e -e i i hyperconjugation i c h t

Q 8.19

e ch ch c h h c ch solution h d ch ch c h i ch ch c ch c p c h t 8.7.10 ch types of Organic reactions and mechanisms o Fig. 8.4(b) s p p p s a e r t o c h y u ii 8.8 methOds OF PuriFicatiOn OF Organic cOmPOunds o t s c d d c m t n organic chemistry - some basic principles and techniques o t 8.8.1 sublimation y t t adaptor 8.8.2 crystallisation t i t adaptor conical flask t o t i Fig.8.5 s p s s p s s s i s Fractional Distillation: i r t 8.8.3 distillation t i t t l t c t t b theoretical plate. c o crude oil in petroleum industry. Distillation under reduced pressure: t s conical flask a t Fig.8.6 s p p p g spent-lye in soap industry s s s s a s t t o a e Fig.8.7 p s s organic chemistry - some basic principles and techniques Fig.8.8 s p p ss p ss s p s p ss 8.8.4 Steam Distillation: t differential extraction i t t t t i d p p p s p p p p p i p extraction t t continuous i 8.8.5 chromatography c a a - p t Fig.8.9 s p s s s s p s g s stopper i a t t b t a p Fig.8.10 p sp s s a) Adsorption Chromatography: a c c t organic chemistry - some basic principles and techniques t Column Chromatography: c a t t t rf value retardation factor a s d s s s s s s t Fig.8.12 (a) Fig.8.11 p p p s s s p p Fig.8.12 (b) s p t Thin Layer Chromatography: t tlc tlc t a a t p p s p t s tlc t Partition Chromatography: p p i 8.9 qualitatiVe analysis OF Organic cOmPOunds t c i a t t t 8.9.1 detection of carbon and hydrogen c ii c t t c c o h c o co c oh h o c so c c co ho c co c so b ho ho 8.9.2 detection of Other elements n "lassaigne's test" t n c n s n n n cn n s n c b i c n s c t Fig.8.13 p p s p s p p p

(A) Test for Nitrogen t ii organic chemistry - some basic principles and techniques t p s ii o ii iii -c b i i p -> a ii ii iii cn- cn - -> a - ii cn l t - cn - ho p

(B) Test for Sulphur t

(D) Test for Phosphorus t t t a a s - - -> p s b p n po h po o hno - -> h po nh m o n no hno - -> a nh cn no - s- - -> cn nos - po a m o nh no ho 8.10 quantitatiVe analysis i q i i p n c n scn- s - -> - -> y n scn scn b u i t t n scn n - -> (C) Test for Halogens t n cn n s 8.10.1 carbon and hydrogen b a ii a ch o - -> c co ho Fig.8.14 s s s p ss s s s s sp s t u c 8.10.2 nitrogen t d u

(i) Dumas method: t t t c o - -> chn co l ho n c t 12 x m2 x 100 44 x m 2 x m1 x 100 18 x m p p t n

Q 8.20

o l l d solution r Volume of nitrogen at STP = 12 x 0.198 x 100 44 x 0.246 = 21.95% (Let it be V mL) Percentage of carbon = p PV 1 1 x 273 760 x T1 p p 2 x 0.1014 x 100 18 x 0.246 = 4.58% t = p p a -a ln stp organic chemistry - some basic principles and techniques Fig. 8.15 s pp p ss s s V mL N 2 at STP weighs = p p s p s s s s s s s s 28 x V g 22400 28 x V x 100 22400 x m s 41.9 mL of nitrogen weighs = 28 x 41.9 g 22400 28 x 41.9 x 100 22400 x 0.3 = 17.46% Percentage of nitrogen =

Q 8.21

i d l

(ii) Kjeldahl's method: t c n a t solution t l a Volume of nitrogen at STP l n 273 x 700 x 50 300 x 760 41.9 mL stp t i t Fig.8.16 p s p s s s s s 14 x M x 2(V - V1 / 2) 100 x 1000 m 1.4 x M x 2 (V - V / 2) = m Percentage of N = o - -> nh h so so Na2SO4 + 2NH3 + 2H2O nh h so - -> nh so l h so l n oh l m m h so m n oh l h so l l h so nh l

Q 8.22

m d m h so m l m l l l nh m nh n nh m h so m solution m l 14 x M x 2(V - V1 / 2) gN 1000 l l h so m m l nh m organic chemistry - some basic principles and techniques p 14 x 20 1000 = 14x20x100 Percentage of nitrogen = = 56.0% 1000x0.5 atomic mass of X x m1g molecular mass of AgX

Q 8.23

8.10.3 halogens Carius method: a i c a b c solution m a b a b 80 x 0.12 g a b Percentage of bromine = 80 x 0.12 x 100 188x0.15 = 34.04% 8.10.4 sulphur a c s i t Fig. 8.17 t s p s p s s l c b so t a l m a a a m = atomic mass of X x m1g molecular mass of AgX i b so 32 x m1 x 100 b so 233 x m 32 x m1 x 100 Percentage of sulphur = 233 x m

Q 8.24

i h solution a m b so t b so 32 x 0.4813 g b so 32 x 0.4813 x 100 Percentage of sulphur = 233 x 0.157 = 42.10% t io heat --- -> o c c 1373 K o ---- -> co x io co - -> i co o 8.10.5 Phosphorus a a x b co a co b a b i nh po m o t a m nh po l m m po therefore l m nh po m o = p i 32 x m1 g O2 88 31 x m1 x 100 % 1877 x m m po 62 x m1 x 100 = x% p therefore p 32 x m1 x 100 % 88 x m t p t m po m po chn s t m po 8.10.6 Oxygen t a organic chemistry - some basic principles and techniques summary i sp sp sp t orbitals hybridisation t sp sp sp t a sp c-h s o t sp - sp p o t wedge dash o a functional group t i u p a iuPac c i iupac o t a carbocations r a free radicals heterolytic carbanions t electrophile t nucleophile electromeric homolytic inductive resonance hyperconjugation effects substitution, addition, o elimination p rearrangement t chromatography i a p a lassaigne's test c n dumas n s Kjeldahl's carius t e ercises ch c o ch ch ch ch co ch chcn c h i ch ch ch c ch c ch ch no hconhch i g d h iupac c chch oh iupac d d t t c c b b d h-cooh ch coch h-ch ch g t h h i o nch ch o- ch ch o- e d s c h oh c h no ch ch chcho c h -cho c h -ch ch ch ch c h e i ch cooh - HO -> ch coo- h o organic chemistry - some basic principles and techniques - CN -> ch ch coch ch c cn oh + CH3CO -> c h coch c - hs -> ch ch sh ch ch b ch c ch ch b ch hci -> ch ch - ho -> ch c- ch oh hb b - cic - ch ch -> ch ho b - cb ch ch ch ho a i e i c ccooh e c chcooh ch ch cooh ch c ch cooh chcooh ch c cooh g c d c d s d l d d d e e n e a c cc g a c a m h so t t l m n oh c c i c i ch sp - sp sp - sp i ch - ch - ch - c c -c ch sp - sp sp - sp l p n cn ch c ch cn ch c cn cn ch ch ch ch ch ch ch s c t c d t ch ch i oh -> ch ch oh i

Unit 9: Hydrocarbons

6 exercises

Q 9.1

How do you account for the formation of ethane during chlorination of methane ? Write IUPAC names of the following compounds : (b) CH2=CH-CC-CH3

(a) CH3CH=C(CH3)2 (c) -CH2-CH2-CH=CH2 (d) (f) CH3(CH2)4 CH (CH2)3 CH3 CH2 -CH (CH3)2 (e)

Q 9.2

(See 9.1)

Q 9.4

(g) CH3 - CH = CH - CH2 - CH = CH - CH - CH2 - CH = CH2 | C2H5 For the following compounds, write structural formulas and IUPAC names for all possible isomers having the number of double or triple bond as indicated :

(b) C5H8 (one triple bond)

(a) C4H8 (one double bond) Write IUPAC names of the products obtained by the ozonolysis of the following compounds :

(i) Pent-2-ene

(ii) 3,4-Dimethylhept-3-ene

(iii) 2-Ethylbut-1-ene

(iv) 1-Phenylbut-1-ene

Q 9.23

An alkene 'A' on ozonolysis gives a mixture of ethanal and pentan-3-one. Write structure and IUPAC name of 'A'. An alkene A contains three C - C eight C - H bonds and one C - C pi bond. 'A' on ozonolysis gives two moles of an aldehyde of molar mass 44 u. Write IUPAC name of 'A'. Propanal and pentan-3-one are the ozonolysis products of an alkene? What is the structural formula of the alkene? Write chemical equations for combustion reaction of the following hydrocarbons:

(i) Butane

(ii) Pentene

(iii) Hexyne

(iv) Toluene Draw the cis and trans structures of hex-2-ene. Which isomer will have higher b.p. and why? Why is benzene extra ordinarily stable though it contains three double bonds? What are the necessary conditions for any system to be aromatic? Explain why the following systems are not aromatic? (ii) (iii) (i) How will you convert benzene into

(i) p-nitrobromobenzene

(ii) m- nitrochlorobenzene

(iii) p - nitrotoluene

(iv) acetophenone? In the alkane H3C - CH2 - C(CH3)2 - CH2 - CH(CH3)2 identif carbon atoms and give the number of H atoms bonded to each one of these. What effect does branching of an alkane chain has on its boiling point? Addition of HBr to propene yields 2-bromopropane, while in the presence of benzoyl peroxide, the same reaction yields 1-bromopropane. Explain and give mechanism. Write down the products of ozonolysis of 1,2-dimethylbenzene (o-xylene). How does the result support Kekul structure for benzene? Arrange benzene, n-hexane and ethyne in decreasing order of acidic behaviour. Also give reason for this behaviour. Why does benzene undergo electrophilic substitution reactions easily and nucleophilic substitutions with difficulty? How would you convert the following compounds into benzene?

(i) Ethyne

(ii) Ethene

(iii) Hexane Write structures of all the alkenes which on hydrogenation give 2-methylbutane. Arrange the following set of compounds in order of their decreasing relative + reactivity with an electrophile, E

(a) Chlorobenzene, 2,4-dinitrochlorobenzene, p-nitrochlorobenzene

(b) Toluene, p-H3C - C6H4 - 2, p-O2N - C6H4 - 2. Out of benzene, m-dinitroben ene and toluene which will undergo nitration most easily and why? Suggest the name of a Lewis acid other than anhydrous aluminium chloride which can be used during ethylation of benzene. Why is Wurtz reaction not preferred for the preparation of alkanes containing odd number of carbon atoms? Illustrate your answer by taking one example.

Q 9.24

(See 9.23)

Q 9.25

(See 9.23)

Class XII Chemistry

Unit 1: Solutions

39 exercises

Q 1.1
Define the term solution. How many types of solutions are formed? Write briefly about each type with an example.

Q 1.2
Give an example of a solid solution in which the solute is a gas.

Q 1.3
Define the following terms:
(i) Mole fraction

Q 1.4
(ii) Molality
(iii) Molarity
(iv) Mass percentage. Concentrated nitric acid used in laboratory work is 68% nitric acid by mass in aqueous solution. What should be the molarity of such a sample of the acid if the density of the solution is 1.504 g mL-1? 27 Solutions

Q 1.5
A solution of glucose in water is labelled as 10% w/w, what would be the molality and mole fraction of each component in the solution? If the density of solution is 1.2 g mL-1, then what shall be the molarity of the solution?

Q 1.6
How many mL of 0.1 M HCl are required to react completely with 1 g mixture of Na2CO3 and NaHCO3 containing equimolar amounts of both?

Q 1.7
A solution is obtained by mixing 300 g of 25% solution and 400 g of 40% solution by mass. Calculate the mass percentage of the resulting solution.

Q 1.8
An antifreeze solution is prepared from 222.6 g of ethylene glycol (C2H6O2) and 200 g of water. Calculate the molality of the solution. If the density of the solution is 1.072 g mL-1, then what shall be the molarity of the solution?

Q 1.9
A sample of drinking water was found to be severely contaminated with chloroform (CHCl3) supposed to be a carcinogen. The level of contamination was 15 ppm (by mass):
(i) express this in percent by mass
(ii) determine the molality of chloroform in the water sample.

Q 1.10
What role does the molecular interaction play in a solution of alcohol and water? 1. 1 1 Why do gases always tend to be less soluble in liquids as the temperature is raised?

Q 1.12
State Henry's law and mention some important applications.

Q 1.13
The partial pressure of ethane over a solution containing 6.56 x 10-3 g of ethane is 1 bar. If the solution contains 5.00 x 10-2 g of ethane, then what shall be the partial pressure of the gas?

Q 1.14
What is meant by positive and negative deviations from Raoult's law and how is the sign of DmixH related to positive and negative deviations from Raoult's law?

Q 1.15
An aqueous solution of 2% non-volatile solute exerts a pressure of 1.004 bar at the normal boiling point of the solvent. What is the molar mass of the solute?

Q 1.16
Heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components are 105.2 kPa and 46.8 kPa respectively. What will be the vapour pressure of a mixture of 26.0 g of heptane and 35 g of octane?

Q 1.17

The vapour pressure of water is 12.3 kPa at 300 K. Calculate vapour pressure of 1 molal solution of a non-volatile solute in it.

Q 1.18

Calculate the mass of a non-volatile solute (molar mass 40 g mol-1) which should be dissolved in 114 g octane to reduce its vapour pressure to 80%.

Q 1.19

A solution containing 30 g of non-volatile solute exactly in 90 g of water has a vapour pressure of 2.8 kPa at 298 K. Further, 18 g of water is then added to the solution and the new vapour pressure becomes 2.9 kPa at 298 K. Calculate:
(i) molar mass of the solute
(ii) vapour pressure of water at 298 K.

Q 1.20

A 5% solution (by mass) of cane sugar in water has freezing point of 271K. Calculate the freezing point of 5% glucose in water if freezing point of pure water is 273.15 K.

Q 1.21

Two elements A and B form compounds having formula AB2 and AB4. When dissolved in 20 g of benzene (C6H6), 1 g of AB2 lowers the freezing point by 2.3 K whereas 1.0 g of AB4 lowers it by 1.3 K. The molar depression constant for benzene is 5.1 K kg mol-1. Calculate atomic masses of A and B. 28

Q 1.22

At 300 K, 36 g of glucose present in a litre of its solution has an osmotic pressure of 4.98 bar. If the osmotic pressure of the solution is 1.52 bars at the same temperature, what would be its concentration?

Q 1.23

Suggest the most important type of intermolecular attractive interaction in the following pairs.
(i) n-hexane and n-octane
(ii) I2 and CCl4
(iii) NaClO4 and water
(iv) methanol and acetone
(v) acetonitrile (CH3CN) and acetone (C3H6O).

Q 1.24

Based on solute-solvent interactions, arrange the following in order of increasing solubility in n-octane and explain. Cyclohexane, KCl, CH3OH, CH3CN.

Q 1.25

Amongst the following compounds, identify which are insoluble, partially soluble and highly soluble in water?
(i) phenol
(iv) ethylene glycol
(ii) toluene
(v) chloroform
(iii) formic acid
(vi) pentanol.

Q 1.26

If the density of some lake water is 1.25g mL-1 and contains 92 g of Na+ ions per kg of water, calculate the molarity of Na+ ions in the lake.

Q 1.27

If the solubility product of CuS is 6 x 10-16, calculate the maximum molarity of CuS in aqueous solution.

Q 1.28

Calculate the mass percentage of aspirin (C9H8O4) in acetonitrile (CH3CN) when 6.5 g of C9H8O4 is dissolved in 450 g of CH3CN.

Q 1.29

Nalorphene (C19H 21NO3), similar to morphine, is used to combat withdrawal symptoms in narcotic users. Dose of nalorphene generally given is 1.5 mg. Calculate the mass of 1.5 10-3 m aqueous solution required for the above dose.

Q 1.30

Calculate the amount of benzoic acid (C6H5COOH) required for preparing 250 mL of 0.15 M solution in methanol.

Q 1.31
The depression in freezing point of water observed for the same amount of acetic acid, trichloroacetic acid and trifluoroacetic acid increases in the order given above. Explain briefly.

Q 1.32
Calculate the depression in the freezing point of water when 10 g of CH3CH2CHClCOOH is added to 250 g of water. K a = 1.4 x 10-3, Kf = 1.86 K kg mol-1.

Q 1.33
19.5 g of CH2FCOOH is dissolved in 500 g of water. The depression in the freezing point of water observed is 1.00 C. Calculate the van't Hoff factor and dissociation constant of fluoroacetic acid.

Q 1.34
Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water.

Q 1.35
Henry's law constant for the molality of methane in benzene at 298 K is 4.27 x 105 mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.

Q 1.36
100 g of liquid A (molar mass 140 g mol-1) was dissolved in 1000 g of liquid B (molar mass 180 g mol-1). The vapour pressure of pure liquid B was found to be 500 torr. Calculate the vapour pressure of pure liquid A and its vapour pressure in the solution if the total vapour pressure of the solution is 475 Torr. 29 Solutions 1. 3 7 Vapour pressures of pure acetone and chloroform at 328 K are 741.8 mm Hg and 632.8 mm Hg respectively. Assuming that they form ideal solution over the entire range of composition, plot p total, p chloroform, and p acetone as a function of xacetone. The experimental data observed for different compositions of mixture is: 100 x xacetone 0 11.8 pacetone /mm Hg 0 54.9 110.1 202.4 322.7 405.9 454.1 521.1 pchloroform /mm Hg 23.4 36.0 50.8 58.2 64.5 72.1 632.8 548.1 469.4 359.7 257.7 193.6 161.2 120.7 Plot this data also on the same graph paper. Indicate whether it has positive deviation or negative deviation from the ideal solution.

Q 1.38
Benzene and toluene form ideal solution over the entire range of composition. The vapour pressure of pure benzene and toluene at 300 K are 50.71 mm Hg and 32.06 mm Hg respectively. Calculate the mole fraction of benzene in vapour phase if 80 g of benzene is mixed with 100 g of toluene.

Q 1.39
The air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298 K if the Henry's law constants for oxygen and nitrogen at 298 K are 3.30 x 107 mm and 6.51 x 107 mm respectively, calculate the composition of these gases in water.

Q 1.40
Determine the amount of CaCl2 (i = 2.47) dissolved in 2.5 litre of water such that its osmotic pressure is 0.75 atm at 27deg C.

Q 1.41
Determine the osmotic pressure of a solution prepared by dissolving 25 mg of K2SO4 in 2 litre of water at 25deg C, assuming that it is completely dissociated.

Unit 2: Electrochemistry

18 exercises

Q 2.1
Arrange the following metals in the order in which they displace each other from the solution of their salts. Al, Cu, Fe, Mg and Zn.

Q 2.2
Given the standard electrode potentials, K+/K = -2.93V, Ag+/Ag = 0.80V, Hg2+/Hg = 0.79V Mg2+/Mg = -2.37 V, Cr3+/Cr = - 0.74V Arrange these metals in their increasing order of reducing power.

Q 2.3
Depict the galvanic cell in which the reaction Zn(s)+2Ag+(aq) (R)Zn2+(aq)+2Ag(s) takes place. Further show:
(i) Which of the electrode is negatively charged?
(ii) The carriers of the current in the cell.
(iii) Individual reaction at each electrode.

Q 2.4

Calculate the standard cell potentials of galvanic cell in which the following reactions take place:

(i) 2Cr(s) + 3Cd2+(aq) 2+ (R) 2Cr3+(aq) + 3Cd + (ii) Fe (aq) + Ag (aq) (R) Fe3+(aq) + Ag(s) Calculate the DrG o and equilibrium constant of the reactions.

Q 2.5

Write the Nernst equation and emf of the following cells at 298 K:

(i) Mg(s)|Mg2+(0.001M)||Cu2+(0.0001 M)|Cu(s)

(ii) Fe(s)|Fe2+(0.001M)||H+(1M)|H2(g)(1bar)| Pt(s)

(iii) Sn(s)|Sn2+(0.050 M)||H+(0.020 M)|H2(g) (1 bar)|Pt(s)

(iv) Pt(s)|Br-(0.010 M)|Br2(l)||H+(0.030 M)| H2(g) (1 bar)|Pt(s).

Q 2.6

In the button cells widely used in watches and other devices the following reaction takes place: Zn(s) + Ag2O(s) + H2O(l) (R) Zn2+(aq) + 2Ag(s) + 2OH-(aq) Determine Dr Go and E o for the reaction.

Q 2.7

Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration.

Q 2.8

The conductivity of 0.20 M solution of KCl at 298 K is 0.0248 S cm-1. Calculate its molar conductivity.

Q 2.9

The resistance of a conductivity cell containing 0.001M KCl solution at 298 K is 1500 W. What is the cell constant if conductivity of 0.001M KCl solution at 298 K is 0.146 x 10-3 S cm-1. 59 Electrochemistry

Q 2.10

The conductivity of sodium chloride at 298 K has been determined at different concentrations and the results are given below: Concentration/M 0.001 0.010 0.020 0.050 -1 11.85 23.15 55.53 106.74 2 10 x k/S m 1.237 0.100 Calculate Lambdam for all concentrations and draw a plot between Lambdam and c1/2. Find the value of L0m .

Q 2.11

Conductivity of 0.00241 M acetic acid is 7.896 x 10-5 S cm-1. Calculate its molar conductivity. If L0m for acetic acid is 390.5 S cm2 mol-1, what is its dissociation constant?

Q 2.12

How much charge is required for the following reductions:

(i) 1 mol of Al3+ to Al ?

(ii) 1 mol of Cu2+ to Cu ?

(iii) 1 mol of MnO4- to Mn2+ ?

Q 2.13

How much electricity in terms of Faraday is required to produce

(i) 20.0 g of Ca from molten CaCl2?

(ii) 40.0 g of Al from molten Al2O3?

Q 2.14

How much electricity is required in coulomb for the oxidation of

(i) 1 mol of H2O to O2?

(ii) 1 mol of FeO to Fe2O3?

Q 2.15

A solution of Ni(NO3)2 is electrolysed between platinum electrodes using a current of 5 amperes for 20 minutes. What mass of Ni is deposited at the cathode?

Q 2.16

Three electrolytic cells A,B,C containing solutions of ZnSO4, AgNO3 and CuSO4, respectively are connected in series. A steady current of 1.5 amperes was passed through them until 1.45 g of silver deposited at the cathode of cell B. How long did the current flow? What mass of copper and zinc were deposited?

Q 2.17

Using the standard electrode potentials given in Table 3.1, predict if the reaction between the following is feasible:

(i) Fe3+(aq) and I-(aq)

(ii) Ag+ (aq) and Cu(s)

(iii) Fe3+ (aq) and Br- (aq)

(iv) Ag(s) and Fe 3+ (aq)

(v) Br 2 (aq) and Fe2+ (aq).

Q 2.18

Predict the products of electrolysis in each of the following:
(i) An aqueous solution of $AgNO_3$ with silver electrodes.
(ii) An aqueous solution of $AgNO_3$ with platinum electrodes.
(iii) A dilute solution of H_2SO_4 with platinum electrodes.
(iv) An aqueous solution of $CuCl_2$ with platinum electrodes.

Unit 3: Chemical Kinetics

13 exercises

Q 3.1

From the rate expression for the following reactions, determine their order of reaction and the dimensions of the rate constants.
(i) 3NO(g) (R) N_2O (g) Rate = k[NO]2
(ii) H_2O_2 (aq) + 3I- (aq) + 2H+ (R) $2H_2O$ (l) + I-3
(iii) CH_3CHO (g) (R) CH_4 (g) + CO(g) Rate = k[H_2O_2][I-] Rate = k [CH_3CHO]3/2
(iv) C_2H_5Cl (g) (R) C_2H_4 (g) + HCl (g) Rate = k [C_2H_5Cl]

Q 3.2

For the reaction: 2A + B (R) A_2B the rate = k[A][B]2 with k = 2.0 x 10-6 mol-2 L2 s-1. Calculate the initial rate of the reaction when [A] = 0.1 mol L-1, [B] = 0.2 mol L-1. Calculate the rate of reaction after [A] is reduced to 0.06 mol L-1.

Q 3.3

The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if k = 2.5 x 10-4 mol-1 L s-1?

Q 3.4

The decomposition of dimethyl ether leads to the formation of CH_4, H_2 and CO and the reaction rate is given by Rate = k [CH_3OCH_3]3/2 The rate of reaction is followed by increase in pressure in a closed vessel, so the rate can also be expressed in terms of the partial pressure of dimethyl ether, i.e., Rate = k (p CH_3 OCH_3) 3/2

Q 3.7

3.8 If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constants? Mention the factors that affect the rate of a chemical reaction. A reaction is second order with respect to a reactant. How is the rate of reaction affected if the concentration of the reactant is
(i) doubled
(ii) reduced to half ? What is the effect of temperature on the rate constant of a reaction? How can this effect of temperature on rate constant be represented quantitatively? In a pseudo first order reaction in water, the following results were obtained: t/s -1 [A]/ mol L

Q 3.9

0 30 60 90 0.55 0.31 0.17 0.085 Calculate the average rate of reaction between the time interval 30 to 60 seconds. A reaction is first order in A and second order in B.
(i) Write the differential rate equation.
(ii) How is the rate affected on increasing the concentration of B three times?
(iii) How is the rate affected when the concentrations of both A and B are doubled? 85 Chemical Kinetics

Q 3.10

In a reaction between A and B, the initial rate of reaction (r0) was measured for different initial concentrations of A and B as given below: A/ mol L-1 0.20 B/ mol L-1 0.30 -1 -1 0.40 0.10 -5 r0/mol L s

Q 3.11

0.20 5.07 x 10 0.05 -5 5.07 x 10 1.43 x 10-4 What is the order of the reaction with respect to A and B? The following results have been obtained during the kinetic studies of the reaction: 2A + B (R) C + D

Q 3.14

3.15 Experiment [A]/mol L-1 [B]/mol L-1 Initial rate of formation of D/mol L-1 min-1 I 0.1 0.1 6.0 x 10-3 II 0.3 0.2 7.2 x 10-2 III 0.3 0.4 2.88 x 10-1 IV 0.4 0.1 2.40 x 10-2 Determine the rate law and the rate constant for the reaction. The reaction between A and B is first order with respect to A and zero order with respect to B. Fill in the blanks in the following table: Experiment [A]/ mol L-1 [B]/ mol L-1 Initial rate/ mol L-1 min-1 I 0.1 0.1 2.0 x 10-2 II - 0.2 4.0 x 10-2 III 0.4 0.4 - IV - 0.2 2.0 x 10-2 Calculate the half-life of a first order reaction from their rate constants given below:

(i) 200 s-1

(ii) 2 min-1

(iii) 4 years-1 The half-life for radioactive decay of 14C is 5730 years. An archaeological artifact containing wood had only 80% of the 14C found in a living tree. Estimate the age of the sample. The experimental data for decomposition of N2O5 [2N2O5 (R) 4NO2 + O2] in gas phase at 318K are given below: t/s 2 0 10 x [N2O5]/ 1.63 mol L-1 1200 1600 2000 2400 2800 3200 1.36 1.14 0.93 0.78

(i) Plot [N2O5] against t.

(ii) Find the half-life period for the reaction.

(iii) Draw a graph between log[N2O5] and t.

(iv) What is the rate law ? 86 0.64 0.53 0.43 0.35

Q 3.20

3.21

(v) Calculate the rate constant.

(vi) Calculate the half-life period from k and compare it with (ii). The rate constant for a first order reaction is 60 s-1. How much time will it take to reduce the initial concentration of the reactant to its 1/16th value? During nuclear explosion, one of the products is 90Sr with half-life of 28.1 years. If 1mg of 90Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically. For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction. A first order reaction takes 40 min for 30% decomposition. Calculate t1/2. For the decomposition of azoisopropane to hexane and nitrogen at 543 K, the following data are obtained. t (sec) P(mm of Hg) 0 35.0 54.0 63.0 Calculate the rate constant. The following data were obtained during the first order thermal decomposition of SO2Cl2 at a constant volume. SO2 Cl2 (g) -> SO2 (g) + Cl 2 (g) Experiment Time/s-1 Total pressure/atm 1 0 0.5 2 0.6 Calculate the rate of the reaction when total pressure is 0.65 atm.

Q 3.22

The rate constant for the decomposition of N2O5 at various temperatures is given below: T/degC 5 -1 10 x k/s

Q 3.25

3.26 0 20 40 60 80 0.0787 1.70 25.7 2140 Draw a graph between ln k and 1/T and calculate the values of A and Ea. Predict the rate constant at 30deg and 50degC. The rate constant for the decomposition of hydrocarbons is 2.418 x 10-5s-1 at 546 K. If the energy of activation is 179.9 kJ/mol, what will be the value of pre-exponential factor. Consider a certain reaction A (R) Products with k = 2.0 x 10 -2s-1. Calculate the concentration of A remaining after 100 s if the initial concentration of A is 1.0 mol L-1. Sucrose decomposes in acid solution into glucose and fructose according to the first order rate law, with t1/2 = 3.00 hours. What fraction of sample of sucrose remains after 8 hours ? The decomposition of hydrocarbon follows the equation k = (4.5 x 1011s-1) e-28000K/T Calculate Ea. 87 Chemical Kinetics

Q 3.29

3.30 The rate constant for the first order decomposition of H2O2 is given by the following equation: log k = 14.34 - 1.25 x 104K/T Calculate Ea for this reaction and at what temperature will its half-period be 256 minutes? The decomposition of A into product has value of k as 4.5 x 103 s-1 at 10degC and energy of activation 60 kJ mol-1. At what temperature would k be 1.5 x 104s-1? The time required for 10% completion of a first order reaction at 298K is equal to that required for its 25% completion at 308K. If the value of A is 4 x 1010s-1. Calculate k at 318K and Ea. The rate of a reaction quadruples when the temperature changes from 293 K to 313 K. Calculate the energy of activation of the reaction assuming that it does not change with temperature.

Unit 4: The d- and f-Block Elements

38 exercises

Q 4.1

Write down the electronic configuration of:
(i) $Cr3+$
(iii) $Cu+$
(v) $Co2+$ 3+ 4+
(ii) Pm
(iv) Ce
(vi) $Lu2+$ 2+ (vii) $Mn2+$
(viii) $Th4+$ 2+

Q 4.2

Why are Mn +3 state?

Q 4.3

Explain briefly how +2 state becomes more and more stable in the first half of the first row transition elements with increasing atomic number?

Q 4.4

To what extent do the electronic configurations decide the stability of oxidation states in the first series of the transition elements? Illustrate your answer with examples.

Q 4.5

What may be the stable oxidation state of the transition element with the 3 following d electron configurations in the ground state of their atoms : 3d , 5 8 4 3d , 3d and 3d ?

Q 4.6

Name the oxometal anions of the first series of the transition metals in which the metal exhibits the oxidation state equal to its group number.

Q 4.7

What is lanthanoid contraction? What are the consequences of lanthanoid contraction?

Q 4.8

What are the characteristics of the transition elements and why are they called transition elements? Which of the d-block elements may not be regarded as the transition elements?

Q 4.9

In what way is the electronic configuration of the transition elements different from that of the non transition elements? compounds more stable than Fe towards oxidation to their

Q 4.10

What are the different oxidation states exhibited by the lanthanoids?

Q 4.11

Explain giving reasons:
(i) Transition metals and many of their compounds show paramagnetic behaviour.
(ii) The enthalpies of atomisation of the transition metals are high.
(iii) The transition metals generally form coloured compounds.
(iv) Transition metals and their many compounds act as good catalyst.

Q 4.12

What are interstitial compounds? Why are such compounds well known for transition metals?

Q 4.13

How is the variability in oxidation states of transition metals different from that of the non transition metals? Illustrate with examples.

Q 4.14

Describe the preparation of potassium dichromate from iron chromite ore. What is the effect of increasing pH on a solution of potassium dichromate?

Q 4.15

Describe the oxidising action of potassium dichromate and write the ionic equations for its reaction with:
(i) iodide
(ii) iron(II) solution and
(iii) $H2S$ 115 The d- and f- Block Elements

Q 4.16
Describe the preparation of potassium permanganate. How does the acidified permanganate solution react with (i) iron(II) ions (ii) SO2 and (iii) oxalic acid? Write the ionic equations for the reactions.

Q 4.17
For M /M and M /M systems the E o values for some metals are as follows: 2+ 3 2+ Cr /Cr -0.9V Cr /Cr -0.4 V 2+ 3+ Mn /Mn -1.2V Mn /Mn2+ +1.5 V 2+ 3+ 2+ Fe /Fe -0.4V Fe /Fe +0.8 V Use this data to comment upon:
(i) the stability of Fe3+ in acid solution as compared to that of Cr3+ or Mn3+ and
(ii) the ease with which iron can be oxidised as compared to a similar process for either chromium or manganese metal.

Q 4.18
Predict which of the following will be coloured in aqueous solution? Ti , V , + 3+ 2+ 3+ 2+ Cu , Sc , Mn , Fe and Co . Give reasons for each.

Q 4.19
Compare the stability of +2 oxidation state for the elements of the first transition series.

Q 4.20
Compare the chemistry of actinoids with that of the lanthanoids with special reference to:
(i) electronic configuration
(iii) oxidation state
(ii) atomic and ionic sizes and
(iv) chemical reactivity.

Q 4.21
How would you account for the following:
(i) Of the d4 species, Cr2+ is strongly reducing while manganese(III) is strongly oxidising.
(ii) Cobalt(II) is stable in aqueous solution but in the presence of complexing reagents it is easily oxidised.
(iii) The d1 configuration is very unstable in ions.

Q 4.22
What is meant by 'disproportionation'? Give two examples of disproportionation reaction in aqueous solution.

Q 4.23
Which metal in the first series of transition metals exhibits +1 oxidation state most frequently and why?

Q 4.24
Calculate the number of unpaired electrons in the following gaseous ions: Mn , 3+ 3+ 3+ Cr , V and Ti . Which one of these is the most stable in aqueous solution?

Q 4.25
Give examples and suggest reasons for the following features of the transition metal chemistry:
(i) The lowest oxide of transition metal is basic, the highest is amphoteric/acidic.
(ii) A transition metal exhibits highest oxidation state in oxides and fluorides.
(iii) The highest oxidation state is exhibited in oxoanions of a metal.

Q 4.26
Indicate the steps in the preparation of:
(i) K2Cr2O7 from chromite ore.
(ii) KMnO4 from pyrolusite ore.

Q 4.27
What are alloys? Name an important alloy which contains some of the lanthanoid metals. Mention its uses.

Q 4.28
What are inner transition elements? Decide which of the following atomic numbers are the atomic numbers of the inner transition elements : 29, 59, 74, 95, 102, 104.

Q 4.29
The chemistry of the actinoid elements is not so smooth as that of the lanthanoids. Justify this statement by giving some examples from the oxidation state of these elements.

Q 4.30
Which is the last element in the series of the actinoids? Write the electronic configuration of this element. Comment on the possible oxidation state of this element. 2+ 3+ 2+ 3+ 3+ 3+

Q 4.31
Use Hund's rule to derive the electronic configuration of Ce3+ ion, and calculate its magnetic moment on the basis of 'spin-only' formula.

Q 4.32
Name the members of the lanthanoid series which exhibit +4 oxidation states and those which exhibit +2 oxidation states. Try to correlate this type of behaviour with the electronic configurations of these elements.

Q 4.33
Compare the chemistry of the actinoids with that of lanthanoids with reference to:
(i) electronic configuration
(ii) oxidation states and
(iii) chemical reactivity.

Q 4.34
Write the electronic configurations of the elements with the atomic numbers 61, 91, 101, and 109.

Q 4.35
Compare the general characteristics of the first series of the transition metals with those of the second and third series metals in the respective vertical columns. Give special emphasis on the following points:
(i) electronic configurations and (iv) atomic sizes.
(ii) oxidation states
(iii) ionisation enthalpies

Q 4.36
Write down the number of 3d electrons in each of the following ions: Ti2+, V2+, Cr3+, Mn2+, Fe2+, Fe3+, Co2+, Ni2+ and Cu2+. Indicate how would you expect the five 3d orbitals to be occupied for these hydrated ions (octahedral).

Q 4.37
Comment on the statement that elements of the first transition series possess many properties different from those of heavier transition elements.

Q 4.38
What can be inferred from the magnetic moment values of the following complex species ? Example Magnetic Moment (BM) K4[Mn(CN)6) 2+ [Fe(H2O)6] K2[MnCl4] 2.2 5.3 5.9

Unit 5: Coordination Compounds

31 exercises

Q 5.1
Explain the bonding in coordination compounds in terms of Werner's postulates.

Q 5.2
FeSO4 solution mixed with (NH4)2SO4 solution in 1:1 molar ratio gives the 2+ test of Fe ion but CuSO4 solution mixed with aqueous ammonia in 1:4 2+ molar ratio does not give the test of Cu ion. Explain why?

Q 5.3
Explain with two examples each of the following: coordination entity, ligand, coordination number, coordination polyhedron, homoleptic and heteroleptic.

Q 5.4
What is meant by unidentate, didentate and ambidentate ligands? Give two examples for each.

Q 5.5
Specify the oxidation numbers of the metals in the following coordination entities:
(i) [Co(H2O)(CN)(en)2]2+
(iii) [PtCl4]2-
(v) [Cr(NH3)3Cl3] + (ii) [CoBr2(en)2]
(iv) K3[Fe(CN)6]

Q 5.6
Using IUPAC norms write the formulas for the following:
(i) Tetrahydroxidozincate(II)
(vi) Hexaamminecobalt(III) sulphate
(ii) Potassium tetrachloridopalladate(II) (vii) Potassium tri(oxalato)chromate(III)
(iii) Diamminedichloridoplatinum(II)
(viii) Hexaammineplatinum(IV)
(iv) Potassium tetracyanidonickelate(II) (ix) Tetrabromidocuprate(II)
(v) Pentaamminenitrito-O-cobalt(III) (x) Pentaamminenitrito-N-cobalt(III)

Q 5.7
Using IUPAC norms write the systematic names of the following:
(i) [Co(NH3)6]Cl3
(ii) [Pt(NH3)2Cl(NH2CH3)]Cl 3+ (iii) [Ti(H2O)6]
(iv) [Co(NH3)4Cl(NO2)]Cl 2+ (v) [Mn(H2O)6] 2- (vi) [NiCl4]
(vii) [Ni(NH3)6]Cl2 3+ (viii) [Co(en)3]
(ix) [Ni(CO)4]

Q 5.8
List various types of isomerism possible for coordination compounds, giving an example of each.

Q 5.9
How many geometrical isomers are possible in the following coordination entities? 3- (i) [Cr(C2O4)3]
(ii) [Co(NH3)3Cl3]

Q 5.10
Draw the structures of optical isomers of:
(i) [Cr(C2O4)3] 3- (ii) [PtCl2(en)2] 2+ (iii) [Cr(NH3)2Cl2(en)] +

Q 5.11
Draw all the isomers (geometrical and optical) of: + 2+
(i) [CoCl2(en)2]
(ii) [Co(NH3)Cl(en)2] + (iii) [Co(NH3)2Cl2(en)]

Q 5.12
Write all the geometrical isomers of [Pt(NH3)(Br)(Cl)(py)] and how many of these will exhibit optical isomers?

Q 5.13
Aqueous copper sulphate solution (blue in colour) gives:
(i) a green precipitate with aqueous potassium fluoride and
(ii) a bright green solution with aqueous potassium chloride. Explain these experimental results.

Q 5.14
What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate? Why is it that no precipitate of copper sulphide is obtained when H2S(g) is passed through this solution?

Q 5.15
Discuss the nature of bonding in the following coordination entities on the basis of valence bond theory:
(i) [Fe(CN)6] 4- (ii) [FeF6] 3- 3-
(iii) [Co(C2O4)3]
(iv) [CoF6] 3-

Q 5.16
Draw figure to show the splitting of d orbitals in an octahedral crystal field.

Q 5.17
What is spectrochemical series? Explain the difference between a weak field ligand and a strong field ligand.

Q 5.18
What is crystal field splitting energy? How does the magnitude of Do decide the actual configuration of d orbitals in a coordination entity?

Q 5.19
[Cr(NH3)6] 3+ is paramagnetic while [Ni(CN)4] 2+ 2- is diamagnetic. Explain why? 2-

Q 5.20
A solution of [Ni(H2O)6] Explain.

Q 5.21
[Fe(CN)6]

Q 5.22
Discuss the nature of bonding in metal carbonyls.

Q 5.23
Give the oxidation state, d orbital occupation and coordination number of the central metal ion in the following complexes: 4- and [Fe(H2O)6] is green but a solution of [Ni(CN)4] 2+ are of different colours in dilute solutions. Why?

(i) K3[Co(C2O4)3]
(ii) cis-[CrCl2(en)2]Cl

Q 5.24
is colourless.

(iii) (NH4)2[CoF4]
(iv) [Mn(H2O)6]SO4 Write down the IUPAC name for each of the following complexes and indicate the oxidation state, electronic configuration and coordination number. Also give stereochemistry and magnetic moment of the complex:
(i) K[Cr(H2O)2(C2O4)2].3H2O
(iii) [CrCl3(py) 3]
(ii) [Co(NH3)5Cl-]Cl2
(iv) Cs[FeCl 4] 3+ (v) K4[Mn(CN) 6]

Q 5.25
Explain the violet colour of the complex [Ti(H2O)6] field theory.

Q 5.26
What is meant by the chelate effect? Give an example.

Q 5.27
Discuss briefly giving an example in each case the role of coordination compounds in:

(i) biological systems
(ii) medicinal chemistry and

Q 5.28
on the basis of crystal

(iii) analytical chemistry
(iv) extraction/metallurgy of metals. How many ions are produced from the complex Co(NH3)6Cl2 in solution?
(i) 6
(ii) 4
(iii) 3
(iv) 2 Coordination Compounds

Q 5.29
Amongst the following ions which one has the highest magnetic moment value?

(i) [Cr(H2O)6]3+

Q 5.30
(ii) [Fe(H2O)6]2+ (iii) [Zn(H2O)6]2+ Amongst the following, the most stable complex is 3+ (i) [Fe(H2O)6]

Q 5.31
(ii) [Fe(NH3)6] 3+ 3-
(iii) [Fe(C2O4)3]
(iv) [FeCl6] 3- What will be the correct order for the wavelengths of absorption in the visible region for the following: 4- 2+ 2+ [Ni(NO2)6] , [Ni(NH3)6] , [Ni(H2O)6] ?

Unit 6: Haloalkanes and Haloarenes

20 exercises

Q 6.1

6.2 Name the following halides according to IUPAC system and classify them as alkyl, allyl, benzyl (primary, secondary, tertiary), vinyl or aryl halides:
(i) $(CH_3)_2CHCH(Cl)CH_3$
(ii) $CH_3CH_2CH(CH_3)CH(C_2H_5)Cl$
(iii) $CH_3CH_2C(CH_3)_2CH_2I$
(iv) $(CH_3)_3CCH_2CH(Br)C_6H_5$
(v) $CH_3CH(CH_3)CH(Br)CH_3$
(vi) $CH_3C(C_2H_5)_2CH_2Br$
(vii) $CH_3C(Cl)(C_2H_5)CH_2CH_3$
(viii) $CH_3CH=C(Cl)CH_2CH(CH_3)_2$
(ix) $CH_3CH=CHC(Br)(CH_3)_2$
(x) p-$ClC_6H_4CH_2CH(CH_3)_2$
(xi) m-$ClCH_2C_6H_4CH_2C(CH_3)_3$
(xii) o-Br-$C_6H_4CH(CH_3)CH_2CH_3$ Give the IUPAC names of the following compounds:
(i) $CH_3CH(Cl)CH(Br)CH_3$
(iv) $(CCl_3)_3CCl$

Q 6.3

(ii) $CHF_2CBrClF$
(iii) $ClCH_2CCCH_2Br$
(v) $CH_3C(p\text{-}ClC_6H_4)_2CH(Br)CH_3$
(vi) $(CH_3)_3CCH=CClC_6H_4I$-p Write the structures of the following organic halogen compounds.
(i) 2-Chloro-3-methylpentane
(ii) p-Bromochlorobenzene
(iii) 1-Chloro-4-ethylcyclohexane
(iv) 2-(2-Chlorophenyl)-1-iodooctane
(v) 2-Bromobutane
(vi) 4-tert-Butyl-3-iodoheptane
(vii) 1-Bromo-4-sec-butyl-2-methylbenzene
(viii) 1,4-Dibromobut-2-ene

Q 6.4

Which one of the following has the highest dipole moment?

Q 6.5

A hydrocarbon C_5H_{10} does not react with chlorine in dark but gives a single monochloro compound C_5H_9Cl in bright sunlight. Identify the hydrocarbon.

Q 6.6

Write the isomers of the compound having formula C_4H_9Br.

Q 6.7

Write the equations for the preparation of 1-iodobutane from

Q 6.8

What are ambident nucleophiles? Explain with an example.
(i) CH_2Cl 2 (ii) $CHCl_3$
(iii) CCl_4
(i) 1-butanol (ii) 1-chlorobutane
(iii) but-1-ene. 189 Haloalkanes and Haloarenes

Q 6.9

Which compound in each of the following pairs will react faster in SN2 reaction with -OH?
(i) CH_3Br or CH_3I

Q 6.10

(ii) $(CH_3)_3CCl$ or CH_3Cl Predict all the alkenes that would be formed by dehydrohalogenation of the following halides with sodium ethoxide in ethanol and identify the major alkene:
(i) 1-Bromo-1-methylcyclohexane
(ii) 2-Chloro-2-methylbutane
(iii) 2,2,3-Trimethyl-3-bromopentane.

Q 6.11

How will you bring about the following conversions?

(i) Ethanol to but-1-yne (ii) Ethane to bromoethene (iii) Propene to 1-nitropropane (iv) Toluene to benzyl alcohol (v) Propene to propyne

(vi) Ethanol to ethyl fluoride (vii) Bromomethane to propanone (viii) But-1-ene to but-2-ene (ix) 1-Chlorobutane to n-octane (x) Benzene to biphenyl.

Q 6.12

Explain why

(i) the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride?

(ii) alkyl halides, though polar, are immiscible with water?

(iii) Grignard reagents should be prepared under anhydrous conditions?

Q 6.13

Give the uses of freon 12, DDT, carbon tetrachloride and iodoform.

Q 6.14

Write the structure of the major organic product in each of the following reactions:

(i) CH3CH2CH2Cl + NaI

(ii) (CH3)3CBr + KOH

(iii) CH3CH(Br)CH2CH3 + NaOH

(iv) CH3CH2Br + KCN

(v) C6H5ONa + C2H5Cl

(vi) CH3CH2CH2OH + SOCl2

(vii) CH3CH2CH = CH2 + HBr

(viii) CH3CH = C(CH3)2 + HBr

Q 6.15

Write the mechanism of the following reaction: nBuBr

Q 6.16

+ KCN nBuCN Arrange the compounds of each set in order of reactivity towards SN2 displacement:

(i) 2-Bromo-2-methylbutane, 1-Bromopentane, 2-Bromopentane

(ii) 1-Bromo-3-methylbutane, 2-Bromo-2-methylbutane, 2-Bromo-3-methylbutane

(iii) 1-Bromobutane, 1-Bromo-2,2-dimethylpropane, 1-Bromo-2-methylbutane, 1-Bromo-3-methylbutane.

Q 6.17

Out of C6H5CH2Cl and C6H5CHClC6H5, which is more easily hydrolysed by aqueous KOH.

Q 6.18

p-Dichlorobenzene has higher m.p. than those of o- and m-isomers.

Q 6.19

How the following conversions can be carried out?

(i) Propene to propan-1-ol

(ii) Ethanol to but-1-yne

(iii) 1-Bromopropane to 2-bromopropane Discuss.

(iv) Toluene to benzyl alcohol

(v) Benzene to 4-bromonitrobenzene

(vi) Benzyl alcohol to 2-phenylethanoic acid

(vii) Ethanol to propanenitrile

(viii) Aniline to chlorobenzene

(ix) 2-Chlorobutane to 3, 4-dimethylhexane

(x) 2-Methyl-1-propene to 2-chloro-2-methylpropane

(xi) Ethyl chloride to propanoic acid

(xii) But-1-ene to n-butyliodide

(xiii) 2-Chloropropane to 1-propanol

(xiv) Isopropyl alcohol to iodoform

(xv) Chlorobenzene to p-nitrophenol

(xvi) 2-Bromopropane to 1-bromopropane

(xvii) Chloroethane to butane (xviii) Benzene to diphenyl

(xix) tert-Butyl bromide to isobutyl bromide

(xx) Aniline to phenylisocyanide

Q 6.20

6.21 The treatment of alkyl chlorides with aqueous KOH leads to the formation of alcohols but in the presence of alcoholic KOH, alkenes are major products. Explain. Primary alkyl halide C_4H_9Br (a) reacted with alcoholic KOH to give compound (b). Compound (b) is reacted with HBr to give (c) which is an isomer of (a). When

(a) is reacted with sodium metal it gives compound (d), C_8H_{18} which is different from the compound formed when n-butyl bromide is reacted with sodium. Give the structural formula of (a) and write the equations for all the reactions.

Q 6.22

What happens when

(i) n-butyl chloride is treated with alcoholic KOH,

(ii) bromobenzene is treated with Mg in the presence of dry ether,

(iii) chlorobenzene is subjected to hydrolysis,

(iv) ethyl chloride is treated with aqueous KOH,

(v) methyl bromide is treated with sodium in the presence of dry ether,

(vi) methyl chloride is treated with KCN?

Unit 7: Alcohols, Phenols and Ethers

33 exercises

Q 7.1

Write IUPAC names of the following compounds: (i) (ii) (iii) (iv) (v) (vi) (vii) (ix)

(x) C_6H_5-O-C_2H_5

(xi) C_6H_5-O-C_7H_{15}(n-) (xii) (viii)

Q 7.2

Write structures of the compounds whose IUPAC names are as follows:

(i) 2-Methylbutan-2-ol

(ii) 1-Phenylpropan-2-ol

(iii) 3,5-Dimethylhexane -1, 3, 5-triol

(iv) 2,3 - Diethylphenol

(v) 1 - Ethoxypropane

(vi) 2-Ethoxy-3-methylpentane

(vii) Cyclohexylmethanol

(viii) 3-Cyclohexylpentan-3-ol

(ix) Cyclopent-3-en-1-ol

(x) 4-Chloro-3-ethylbutan-1-ol.

Q 7.3

(i) Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names.

(ii) Classify the isomers of alcohols in question 11.3 (i) as primary, secondary and tertiary alcohols.

Q 7.4

Explain why propanol has higher boiling point than that of the hydrocarbon, butane?

Q 7.5

Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact.

Q 7.6

What is meant by hydroboration-oxidation reaction? Illustrate it with an example.

Q 7.7

Give the structures and IUPAC names of monohydric phenols of molecular formula, C_7H_8O.

Q 7.8

While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason.

Q 7.9

Give the equations of reactions for the preparation of phenol from cumene.

Q 7.10

Write chemical reaction for the preparation of phenol from chlorobenzene.

Q 7.11

Write the mechanism of hydration of ethene to yield ethanol.

Q 7.12
You are given benzene, conc. H_2SO_4 and NaOH. Write the equations for the preparation of phenol using these reagents.

Q 7.13
Show how will you synthesise:
(i) 1-phenylethanol from a suitable alkene.
(ii) cyclohexylmethanol using an alkyl halide by an SN2 reaction.
(iii) pentan-1-ol using a suitable alkyl halide?

Q 7.14
Give two reactions that show the acidic nature of phenol. Compare acidity of phenol with that of ethanol.

Q 7.15
Explain why is ortho nitrophenol more acidic than ortho methoxyphenol ?

Q 7.16
Explain how does the -OH group attached to a carbon of benzene ring activate it towards electrophilic substitution?

Q 7.17
Give equations of the following reactions:
(i) Oxidation of propan-1-ol with alkaline $KMnO_4$ solution.
(ii) Bromine in CS_2 with phenol.
(iii) Dilute HNO_3 with phenol.
(iv) Treating phenol wih chloroform in presence of aqueous NaOH.

Q 7.18
Explain the following with an example.
(i) Kolbe's reaction.
(ii) Reimer -Tiemann reaction.
(iii) Williamson ether synthesis.
(iv) Unsymmetrical ether.

Q 7.19
Write the mechanism of acid dehydration of ethanol to yield ethene.

Q 7.20
How are the following conversions carried out?
(i) Propene (R) Propan-2-ol.
(ii) Benzyl chloride (R) Benzyl alcohol.
(iii) Ethyl magnesium chloride (R) Propan-1-ol.
(iv) Methyl magnesium bromide (R) 2-Methylpropan-2-ol.

Q 7.21
Name the reagents used in the following reactions:
(i) Oxidation of a primary alcohol to carboxylic acid.
(ii) Oxidation of a primary alcohol to aldehyde.
(iii) Bromination of phenol to 2,4,6-tribromophenol.
(iv) Benzyl alcohol to benzoic acid.
(v) Dehydration of propan-2-ol to propene.
(vi) Butan-2-one to butan-2-ol.

Q 7.22
Give reason for the higher boiling point of ethanol in comparison to methoxymethane. Alcohols, Phenols and Ethers

Q 7.23
Give IUPAC names of the following ethers:

Q 7.24
Write the names of reagents and equations for the preparation of the following ethers by Williamson's synthesis:
(i) 1-Propoxypropane
(ii) Ethoxybenzene
(iii) 2-Methoxy-2-methylpropane
(iv) 1-Methoxyethane

Q 7.25
Illustrate with examples the limitations of Williamson synthesis for the preparation of certain types of ethers.

Q 7.26
How is 1-propoxypropane synthesised from propan-1-ol? Write mechanism of this reaction.

Q 7.27
Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. Give reason.

Q 7.28
Write the equation of the reaction of hydrogen iodide with:
(i) 1-propoxypropane (ii) methoxybenzene and (iii) benzyl ethyl ether.

Q 7.29
Explain the fact that in aryl alkyl ethers (i) the alkoxy group activates the benzene ring towards electrophilic substitution and (ii) it directs the incoming substituents to ortho and para positions in benzene ring.

Q 7.30
Write the mechanism of the reaction of HI with methoxymethane.

Q 7.31
Write equations of the following reactions:
(i) Friedel-Crafts reaction - alkylation of anisole.
(ii) Nitration of anisole.
(iii) Bromination of anisole in ethanoic acid medium.
(iv) Friedel-Craft's acetylation of anisole.

Q 7.32
Show how would you synthesise the following alcohols from appropriate alkenes? CH3 (i) OH OH (ii) OH (iii) (iv) OH

Q 7.33
When 3-methylbutan-2-ol is treated with HBr, the following reaction takes place: Give a mechanism for this reaction. (Hint : The secondary carbocation formed in step II rearranges to a more stable tertiary carbocation by a hydride ion shift from 3rd carbon atom.

Unit 8: Aldehydes, Ketones and Carboxylic Acids

20 exercises

Q 8.1
What is meant by the following terms ? Give an example of the reaction in each case.
(i) Cyanohydrin
(ii) Acetal
(iii) Semicarbazone
(iv) Aldol
(v) Hemiacetal
(vi) Oxime
(vii) Ketal
(vii) Imine
(ix) 2,4-DNP-derivative
(x) Schiff's base

Q 8.2
Name the following compounds according to IUPAC system of nomenclature:
(i) CH3CH(CH3)CH2CH2CHO
(ii) CH3CH2COCH(C2H5)CH2CH2Cl
(iii) CH3CH=CHCHO
(iv) CH3COCH2COCH3
(v) CH3CH(CH3)CH2C(CH3)2COCH3
(vi) (CH3)3CCH2COOH
(vii) OHCC6H4CHO-p

Q 8.3

Draw the structures of the following compounds.
(i) 3-Methylbutanal
(ii) p-Nitropropiophenone
(iii) p-Methylbenzaldehyde
(iv) 4-Methylpent-3-en-2-one
(v) 4-Chloropentan-2-one
(vi) 3-Bromo-4-phenylpentanoic acid
(vii) p,p'-Dihydroxybenzophenone
(viii) Hex-2-en-4-ynoic acid

Q 8.4

Write the IUPAC names of the following ketones and aldehydes. Wherever possible, give also common names.
(i) CH3CO(CH2)4CH3
(ii) CH3CH2CHBrCH2CH(CH3)CHO
(iii) CH3(CH2)5CHO
(iv) Ph-CH=CH-CHO CHO (v)

Q 8.5

(vi) PhCOPh Draw structures of the following derivatives.
(i) The 2,4-dinitrophenylhydrazone of benzaldehyde
(ii) Cyclopropanone oxime
(iii) Acetaldehydedimethylacetal
(iv) The semicarbazone of cyclobutanone
(v) The ethylene ketal of hexan-3-one
(vi) The methyl hemiacetal of formaldehyde Aldehydes, Ketones and Carboxylic Acids

Q 8.6

Predict the products formed when cyclohexanecarbaldehyde reacts with following reagents. + (i) PhMgBr and then H3O
(ii) Tollens' reagent
(iii) Semicarbazide and weak acid
(iv) Excess ethanol and acid
(v) Zinc amalgam and dilute hydrochloric acid

Q 8.7

Which of the following compounds would undergo aldol condensation, which the Cannizzaro reaction and which neither? Write the structures of the expected products of aldol condensation and Cannizzaro reaction.
(i) Methanal
(ii) 2-Methylpentanal
(iii) Benzaldehyde
(iv) Benzophenone
(v) Cyclohexanone
(vi) 1-Phenylpropanone
(vii) Phenylacetaldehyde (viii) Butan-1-ol
(ix) 2,2-Dimethylbutanal

Q 8.8

How will you convert ethanal into the following compounds?
(i) Butane-1,3-diol
(ii) But-2-enal
(iii) But-2-enoic acid

Q 8.9

Write structural formulas and names of four possible aldol condensation products from propanal and butanal. In each case, indicate which aldehyde acts as nucleophile and which as electrophile.

Q 8.10

An organic compound with the molecular formula C9H10O forms 2,4-DNP derivative, reduces Tollens' reagent and undergoes Cannizzaro reaction. On vigorous oxidation, it gives 1,2-benzenedicarboxylic acid. Identify the compound.

Q 8.11

An organic compound (A) (molecular formula C8H 16O 2) was hydrolysed with dilute sulphuric acid to give a carboxylic acid (B) and an alcohol (C). Oxidation of (C) with chromic acid produced (B). (C) on dehydration gives but-1-ene. Write equations for the reactions involved.

Q 8.12

Arrange the following compounds in increasing order of their property as indicated:

(i) Acetaldehyde, Acetone, Di-tert-butyl ketone, Methyl tert-butyl ketone (reactivity towards HCN)

(ii) CH3CH2CH(Br)COOH, CH3CH(Br)CH2COOH, (CH3)2CHCOOH, CH3CH2CH2COOH (acid strength)

(iii) Benzoic acid, 4-Nitrobenzoic acid, 3,4-Dinitrobenzoic acid, 4-Methoxybenzoic acid (acid strength)

Q 8.13

Give simple chemical tests to distinguish between the following pairs of compounds.

(i) Propanal and Propanone

(ii) Acetophenone and Benzophenone

(iii) Phenol and Benzoic acid

(iv) Benzoic acid and Ethyl benzoate

(v) Pentan-2-one and Pentan-3-one (vi) Benzaldehyde and Acetophenone

(vii) Ethanal and Propanal

Q 8.14

How will you prepare the following compounds from benzene? You may use any inorganic reagent and any organic reagent having not more than one carbon atom

(i) Methyl benzoate

(ii) m-Nitrobenzoic acid

(iii) p-Nitrobenzoic acid

(iv) Phenylacetic acid

(v) p-Nitrobenzaldehyde.

Q 8.15

How will you bring about the following conversions in not more than two steps?

(i) Propanone to Propene

(ii) Benzoic acid to Benzaldehyde

(iii) Ethanol to 3-Hydroxybutanal

(iv) Benzene to m-Nitroacetophenone

(v) Benzaldehyde to Benzophenone

(vi) Bromobenzene to 1-Phenylethanol

(vii) Benzaldehyde to 3-Phenylpropan-1-ol

(viii) Benazaldehyde to a-Hydroxyphenylacetic acid

(ix) Benzoic acid to m- Nitrobenzyl alcohol

Q 8.16

Describe the following:

(i) Acetylation

(ii) Cannizzaro reaction

(iii) Cross aldol condensation

(iv) Decarboxylation

Q 8.17

Complete each synthesis by giving missing starting material, reagent or products

Q 8.18

Give plausible explanation for each of the following:

(i) Cyclohexanone forms cyanohydrin in good yield but 2,2,6-trimethylcyclohexanone does not.

(ii) There are two -NH2 groups in semicarbazide. However, only one is involved in the formation of semicarbazones.

(iii) During the preparation of esters from a carboxylic acid and an alcohol in the presence of an acid catalyst, the water or the ester should be removed as soon as it is formed.

Q 8.19

An organic compound contains 69.77% carbon, 11.63% hydrogen and rest oxygen. The molecular mass of the compound is 86. It does not reduce Tollens' reagent but forms an addition compound with sodium hydrogensulphite and give positive iodoform test. On vigorous oxidation it gives ethanoic and propanoic acid. Write the possible structure of the compound.

Q 8.20
Although phenoxide ion has more number of resonating structures than carboxylate ion, carboxylic acid is a stronger acid than phenol. Why?

Unit 9: Amines

11 exercises

Q 9.1
Write IUPAC names of the following compounds and classify them into primary, secondary and tertiary amines.
(i) $(CH_3)_2CHNH_2$
(ii) $CH_3(CH_2)_2NH_2$
(iii) $CH_3NHCH(CH_3)_2$
(iv) $(CH_3)_3CNH_2$
(v) $C_6H_5NHCH_3$
(vi) $(CH_3CH_2)_2NCH_3$
(vii) m-$BrC_6H_4NH_2$

Q 9.2
Give one chemical test to distinguish between the following pairs of compounds.
(i) Methylamine and dimethylamine
(ii) Secondary and tertiary amines
(iii) Ethylamine and aniline
(iv) Aniline and benzylamine
(v) Aniline and N-methylaniline.

Q 9.3
Account for the following:
(i) pKb of aniline is more than that of methylamine.
(ii) Ethylamine is soluble in water whereas aniline is not.
(iii) Methylamine in water reacts with ferric chloride to precipitate hydrated ferric oxide.
(iv) Although amino group is o- and p- directing in aromatic electrophilic substitution reactions, aniline on nitration gives a substantial amount of m-nitroaniline.
(v) Aniline does not undergo Friedel-Crafts reaction.
(vi) Diazonium salts of aromatic amines are more stable than those of aliphatic amines.
(vii) Gabriel phthalimide synthesis is preferred for synthesising primary amines.

Q 9.4
Arrange the following:
(i) In decreasing order of the pKb values: $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_2H_5)_2NH$ and $C_6H_5NH_2$
(ii) In increasing order of basic strength: $C_6H_5NH_2$, $C_6H_5N(CH_3)_2$, $(C_2H_5)_2NH$ and CH_3NH_2
(iii) In increasing order of basic strength:
(a) Aniline, p-nitroaniline and p-toluidine
(b) $C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5CH_2NH_2$.
(iv) In decreasing order of basic strength in gas phase: $C_2H_5NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$ and NH_3
(v) In increasing order of boiling point: C_2H_5OH, $(CH_3)_2NH$, $C_2H_5NH_2$
(vi) In increasing order of solubility in water: $C_6H_5NH_2$, $(C_2H_5)_2NH$, $C_2H_5NH_2$.

Q 9.5
How will you convert:
(i) Ethanoic acid into methanamine
(ii) Hexanenitrile into 1-aminopentane
(iii) Methanol to ethanoic acid
(iv) Ethanamine into methanamine
(v) Ethanoic acid into propanoic acid
(vi) Methanamine into ethanamine
(vii) Nitromethane into dimethylamine
(viii) Propanoic acid into ethanoic acid?

Q 9.6
Describe a method for the identification of primary, secondary and tertiary amines. Also write chemical equations of the reactions involved.

Q 9.7
Write short notes on the following:
(i) Carbylamine reaction
(ii) Diazotisation
(iii) Hofmann's bromamide reaction
(iv) Coupling reaction
(v) Ammonolysis
(vi) Acetylation
(vii) Gabriel phthalimide synthesis.

Q 9.8
Accomplish the following conversions:
(i) Nitrobenzene to benzoic acid
(ii) Benzene to m-bromophenol
(iii) Benzoic acid to aniline
(iv) Aniline to 2,4,6-tribromofluorobenzene
(v) Benzyl chloride to 2-phenylethanamine
(vi) Chlorobenzene to p-chloroaniline
(vii) Aniline to p-bromoaniline
(viii) Benzamide to toluene
(ix) Aniline to benzyl alcohol.

Q 9.9
Give the structures of A, B and C in the following reactions: OH- NaOH + Br 2 -> A ------- -> B ----- -> C
(i) CH3 CH2 I ---- Partial hydrolysis NaCN + CuCN H2 O / H NH3
(ii) C6 H5 N 2 Cl ---- -> A ----- -> B --- -> C D LiAlH HNO 4 2 -> B ---- -> C
(iii) CH3 CH2 Br --- -> A ---- 0C KCN NaNO + HCl H O / H+ 2 2 -> B ----- -> C
(iv) C6 H5 NO2 ---- -> A ------ 273 K D Fe / HCl NH NaOBr NaNO / HCl HNO C H OH 3 2 -> A ---- -> B ------ -> C
(v) CH3 COOH --- D 2 6 5 -> B ---- -> C
(vi) C6 H5 NO2 ---- -> A ---- 273 K Fe / HCl 279 Amines

Q 9.10
9.11 An aromatic compound 'A' on treatment with aqueous ammonia and heating forms compound 'B' which on heating with Br2 and KOH forms a compound 'C' of molecular formula C6H7N. Write the structures and IUPAC names of compounds A, B and C. Complete the following reactions:
(i) C6 H5 NH2 + CHCl3 + alc.KOH ->
(ii) C6 H5 N 2 Cl + H 3 PO 2 + H2 O ->
(iii) C6 H5 NH2 + H2 SO 4 (conc.) ->
(iv) C6 H5 N 2Cl + C2 H5OH ->
(v) C6 H5 NH2 + Br2 (aq) ->
(vi) C6 H5 NH2 + (CH 3 CO)2 O -> ( )HBF i 4 -> (vii) C6 H5 N2 Cl ------- (ii) NaNO2 / Cu, D

Q 9.14
Why cannot aromatic primary amines be prepared by Gabriel phthalimide synthesis? Write the reactions of (i) aromatic and (ii) aliphatic primary amines with nitrous acid. Give plausible explanation for each of the following:
(i) Why are amines less acidic than alcohols of comparable molecular masses?
(ii) Why do primary amines have higher boiling point than tertiary amines?
(iii) Why are aliphatic amines stronger bases than aromatic amines?

Unit 10: Biomolecules

20 exercises

Q 10.5
What are monosaccharides? What are reducing sugars? Write two main functions of carbohydrates in plants. Classify the following into monosaccharides and disaccharides. Ribose, 2-deoxyribose, maltose, galactose, fructose and lactose. What do you understand by the term glycosidic linkage?

Q 10.6
10.7 What is glycogen? How is it different from starch? What are the hydrolysis products of
(i) sucrose and
(ii) lactose?

Q 10.8
What is the basic structural difference between starch and cellulose?

Q 10.9
What happens when D-glucose is treated with the following reagents?
(i) HI
(ii) Bromine water
(iii) HNO3

Q 10.10
Enumerate the reactions of D-glucose which cannot be explained by its open chain structure.

Q 10.11
What are essential and non-essential amino acids? Give two examples of each type.

Q 10.12
Define the following as related to proteins
(i) Peptide linkage
(ii) Primary structure
(iii) Denaturation.

Q 10.13
What are the common types of secondary structure of proteins?

Q 10.14
What type of bonding helps in stabilising the a-helix structure of proteins?

Q 10.15
Differentiate between globular and fibrous proteins.

Q 10.16
How do you explain the amphoteric behaviour of amino acids?

Q 10.17
What are enzymes?

Q 10.18
What is the effect of denaturation on the structure of proteins?

Q 10.19
How are vitamins classified? Name the vitamin responsible for the coagulation of blood.

Q 10.20
Why are vitamin A and vitamin C essential to us? Give their important sources.

Q 10.21
What are nucleic acids? Mention their two important functions.

Q 10.22
What is the difference between a nucleoside and a nucleotide?

Q 10.23
The two strands in DNA are not identical but are complementary. Explain.

Q 10.24
Write the important structural and functional differences between DNA and RNA.

Q 10.25
What are the different types of RNA found in the cell?